

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284331

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Raghavulu; Venkat et al.

SYSTEMS, METHODS, AND APPARATUS FOR DISPLAY PANEL HUMAN PRESENCE DETECTION

Abstract

Methods, apparatus, systems and articles of manufacture are disclosed for display panel human presence detection. An example display device includes a sensor to generate a detection signal based on whether a human is detected, and logic circuitry to generate a control signal to cause an adjustment of a power state of the display device based on the detection signal.

Inventors: Raghavulu; Venkat (Bangalore, IN), Han; Ke (Shanghai, CN)

Applicant: Intel Corporation (Santa Clara, CA)

Family ID: 79282722

Assignee: Intel Corporation (Santa Clara, CA)

Appl. No.: 19/184487

Filed: April 21, 2025

Related U.S. Application Data

parent US continuation 17917876 20221007 parent-grant-document US 12306692 WO
continuation PCT/CN2020/098325 20200626 child US 19184487

Publication Classification

Int. Cl.: G06F1/3231 (20190101); G06F1/3296 (20190101); G06F3/14 (20060101)

U.S. Cl.:

CPC G06F1/3231 (20130101); G06F1/3296 (20130101); G06F3/14 (20130101);

Background/Summary

RELATED APPLICATION(S) [0001] This Patent arises from a continuation of U.S. patent application Ser. No. 17/917,876, which was filed on Oct. 7, 2022, as U.S. National Stage Patent Application under 35 U.S.C. § 371 of the PCT Patent Application No. PCT/CN2020/098325, filed Jun. 26, 2020, and entitled “SYSTEMS, METHODS, AND APPARATUS FOR DISPLAY PANEL HUMAN PRESENCE DETECTION.” U.S. patent application Ser. No. 17/917,876, and PCT Patent Application No. PCT/CN2020/098325 is hereby incorporated herein by reference in their entireties. Priority to U.S. patent application Ser. No. 17/917,876, and PCT Patent Application No. PCT/CN2020/098325 is claimed.

FIELD OF THE DISCLOSURE

[0002] This disclosure relates generally to device power management and, more particularly, to systems, methods, and apparatus for display panel human presence detection.

BACKGROUND

[0003] Computing devices can consume relatively large amounts of energy when executing computationally intensive tasks. Power management tools may be deployed to such computing devices to manage energy expenditure and/or extend battery life. Wasteful and/or unnecessary energy consumption by a computing device can unintentionally occur when a human user steps away from the computing device and/or otherwise is no longer present near the computing device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 is a schematic illustration of an example human detection system including an example human detection logic circuit to facilitate power management of an example display panel.

[0005] FIG. 2 is a schematic illustration of another example human detection system to facilitate power management of another example display panel.

[0006] FIG. 3 is a block diagram of host hardware interfacing with a display panel.

[0007] FIG. 4 is a block diagram of an implementation of yet another example human detection system including an example human detection controller to facilitate power management of the example display panel of FIGS. 1 and/or 2.

[0008] FIG. 5A depicts an example truth table.

[0009] FIG. 5B is a schematic illustration of an example control circuit that may implement the example human detection logic circuit of FIG. 1 and/or the example human detection controller of FIG. 4 in accordance with the logic reflected in the example truth table of FIG. 5A.

[0010] FIG. 6 depicts an example logic table reflecting example input and output relationships for example operating conditions associated with the example human detection system of FIG. 2 and/or the example human detection controller of FIG. 4.

[0011] FIG. 7 is a block diagram of example data flow across an example hardware layer and example software layer associated with the example human detection system of FIGS. 1 and/or 2.

[0012] FIG. 8 is a block diagram illustrating data flow across a hardware layer and a software layer associated with the host hardware of FIG. 3.

[0013] FIG. 9 is an example data flow diagram associated with the example human detection system of FIG. 1.

[0014] FIGS. 10A-10B depict an example data flow diagram associated with the example human detection system of FIG. 2.

[0015] FIGS. 11A-11B depict an example data flow diagram associated with the host hardware and the display panel of FIG. 3.

[0016] FIG. 12 is a flowchart representative of example machine readable instructions that may be executed to implement the example human detection controller of FIG. 4, and/or, more generally, the example human detection system of FIG. 4, to invoke example hardware in the example display device of FIG. 1 to adjust a power state of the display device.

[0017] FIG. 13 is a flowchart representative of example machine readable instructions that may be executed to implement the example human detection controller of FIG. 4, and/or, more generally, the example human detection system of FIG. 4, to invoke example hardware in the example display device of FIG. 1 and/or the host hardware of FIGS. 1 and/or 2 to adjust a power state of the display device.

[0018] FIG. 14 is a flowchart representative of example machine readable instructions that may be executed to implement the example human detection controller of FIG. 4, and/or, more generally, the example human detection system of FIG. 4, to invoke host hardware to adjust a power state of the example display device of FIG. 2.

[0019] FIG. 15 is a block diagram of an example processing platform structured to execute the example machine readable instructions of FIGS. 12-14 to implement the example human detection controller of FIG. 4, and/or, more generally, the third example human detection system of FIG. 4.

[0020] FIG. 16 is a block diagram of another example processing platform structured to execute the example machine readable instructions of FIGS. 12-14 to implement the example human detection controller of FIG. 4, and/or, more generally, the third example human detection system of FIG. 4.

[0021] FIG. 17 is a block diagram of yet another example processing platform structured to execute the example machine readable instructions of FIGS. 12-14 to implement the example human detection controller of FIG. 4, and/or, more generally, the third example human detection system of FIG. 4.

[0022] The figures are not to scale. In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts.

[0023] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly that might, for example, otherwise share a same name. As used herein “substantially real time” refers to occurrence in a near instantaneous manner recognizing there may be real world delays for computing time, transmission, etc. Thus, unless otherwise specified, “substantially real time” refers to real time+/-1 second.

DETAILED DESCRIPTION

[0024] Computing devices can consume relatively large amounts of energy when executing computationally intensive tasks. Power management tools may be deployed on computing devices (e.g., on computing devices that are energy limited, such as computing devices that are battery powered). Such power management tools may manage energy expenditure and/or otherwise extend battery life. For example, a host operating system (OS) running on and/or otherwise executed by a central processing unit (CPU) of a computing device may reduce a brightness of a display panel or device to reduce energy consumption. In other examples, the host OS can turn off the display panel when a user (e.g., a human, a human user, etc.) steps away from the computing device and is no longer present in front of the display panel.

[0025] A human presence sensors (HPS) can be used to detect whether a human is proximate to a display panel, such as when the human is in front of the display panel or within a range (e.g., a range of 6 inches to 6 feet, a range of 0 to 5 feet, etc.) and/or angle (e.g., an angle of 5 degrees from a center point of the display, an angle of 30 degrees from the center point, etc.) of the display panel

from which the HPS can detect the presence of the human. For example, the HPS can detect the human in response to the human being in a vicinity of the display panel that is within a threshold range of the HPS. An HPS can be based on one or more different sensing techniques or technologies. For example, an HPS can be an infrared sensor, a time-of-flight laser sensor, an ultrasonic sensor, a millimeter wave radar sensor, or a camera or camera-based sensor. In some instances, the HPS can be coupled to an integrated sensor hub (ISH) on host hardware (e.g., a system-on-a-chip (SoC) including one or more processors). In such instances, the HPS can trigger a wake event (e.g., a wake-on-approach event) in the host OS of the host hardware when the HPS detects human presence. In other instances, the HPS can trigger a sleep event (e.g., a walk-away-lock event) in the host OS when the HPS detects human absence (e.g., the HPS does not detect a presence of a human).

[0026] In some computing devices, the HPS is part of a sensing subsystem in a platform (e.g., an SoC, a motherboard including a processor, etc.). The sensing subsystem can be a sensor hub (e.g., an ISH), or input/output (I/O) interfaces such as an inter-integrated circuit (I2C) protocol interface, a serial peripheral interface (SPI) bus, etc. In such computing devices, the sensing subsystem can communicate to the host OS of the platform about whether absence or presence of a human is detected. The host OS can inform the display subsystem to perform a control task such as to turn on or off the display panel, to reduce display brightness, etc. For instance, the host OS of an SoC can instruct a video display controller (VDC) of the SoC to transmit display data to a display panel when the HPS detects human presence. In other instances, the host OS can instruct the VDC to cease transmitting display data to the display panel when the HPS detects an absence of a human.

[0027] Examples disclosed herein include example human detection systems to invoke example hardware in an example display device to execute human presence detection. Examples disclosed herein reduce and/or eliminate the need for HPS trigger events to be transmitted to, and/or power state tasks (e.g., turning on or off a display, adjusting a backlight brightness or intensity, etc.) to be executed with, an OS, drivers, and/or an I/O sensing subsystem of host hardware (e.g., an SoC). In some disclosed examples, the example hardware is integrated into a display panel to generate HPS trigger events and/or execute power state tasks associated with the display panel. For example, the display panel hardware includes an SoC (e.g., a display SoC), human detection logic (e.g., software, firmware, and/or hardware), I/O interface(s), I/O bus(es), etc., and/or a combination thereof.

[0028] In some disclosed examples, the HPS trigger events are communicated to the display SoC without any host OS-based triggers and/or without any host OS interventions. For example, the host OS-based triggers and/or the host OS interventions correspond to triggers and/or interventions by an OS on a main or host SoC external to the display device. In such examples, the HPS trigger events are communicated from the HPS to the display SoC without going through the main SoC. In some such examples, the human detection systems included in the display panel are fully isolated from the interventions of the main SoC.

[0029] In some disclosed examples, the human detection system includes human detection logic to compare an HPS signal and a connection status signal, such as a hot-plug-detect (HPD) signal. For example, the HPS signal is indicative and/or otherwise representative of human absence or human presence (e.g., the absence of a person in the range of the detection system or the presence of a person within that range). In such examples, the HPS signal is an indication of either human absence or human presence. In some such examples, the HPD signal is indicative and/or otherwise representative of a connection (e.g., an active connection, an enabled connection, etc.) between a display SoC and host hardware (e.g., a main SoC). In such disclosed examples, the HPS signal invokes the display SoC to adjust a power state of the display panel based on whether a human is detected. In some disclosed examples, the human detection logic instructs the host hardware via the HPD signal to trigger transmission of display data to be rendered on the display panel in response to the HPS detecting human presence. In some disclosed examples, the human detection logic

instructs the host hardware to reduce a screen brightness and/or to cease transmission of data to the display in response to the HPS detecting human absence.

[0030] In some disclosed examples, the display SoC obtains the HPS signal and/or detection parameters associated with the HPS from the HPS. In some such disclosed examples, the display SoC transmits detection data including at least one of the HPS signal or the detection parameters to the main SoC. For example, the display SoC transmits the detection data over an auxiliary (AUX) channel of an interface, such as a high definition multimedia interface (HDMI). In some such disclosed examples, the main SoC compares the detection data to a power profile (e.g., a power management profile) associated with at least one of the main SoC or the display SoC. In some examples, the main SoC determines whether to adjust a power state of the display device based on the comparison.

[0031] Advantageously, some examples disclosed herein effectuate human detection in computing systems that do not include an ISH by including the human detection system in the display panel. In some desktop computing systems, the HPS is coupled to host hardware because the distance between the display panel and the host hardware exceeds the capabilities of I2C and general-purpose input/output (GPIO) communications. Advantageously, examples disclosed herein effectuate human detection in computing systems, such as desktop computing systems, notebook or laptop computing systems, gaming systems, television systems, etc., by including the example human detection system in the display SoC of the display panel rather than in the host hardware.

[0032] FIG. 1 is a schematic illustration of a first example computing environment **100** including a first example human detection system **102** to facilitate power management of a first example display panel **104**. The first computing environment **100** includes a first example computing system **106**, which includes the first display panel **104** and an example main system-on-a-chip (SoC) **108**. In this example, the first computing system **106** is a desktop computer, a television (e.g., a smart television, an Internet-enabled television, etc.), a phone (e.g., a smartphone, an Internet-enabled phone, etc.), a gaming system (e.g., a system including a gaming console and a display device), or any other computing device or system. The main SoC **108** may be present, for example, in an example housing **109** of a desktop computing or other device, such as a mobile device, a television, etc.

[0033] The first display panel **104** is a display device including a screen for presenting content. For example, the first display panel **104** is a light emitting diode (LED) display, an organic light emitting diode (OLED) display, a liquid crystal display (LCD), a cathode ray tube (CRT) display, an in-place switching (IPS) display, or a touchscreen. An example display interface **110**, which includes an example receptacle (e.g., an interface receptacle) **112**, couples the first display panel **104** to the main SoC **108**. The display interface **110** of the example of FIG. 1 is a high definition multimedia interface (HDMI) interface. For example, the display interface **110** may include a cable (e.g., an HDMI cable), a first interface port (e.g., a first HDMI interface port) on the first display panel **104**, a second interface port (e.g., a second HDMI interface port) on the main SoC **108** or on the housing **109** of the main SoC **108**, etc. The display interface **110** includes one or more example receptacles **112**. The receptacle **112** of the example of FIG. 1 is an HDMI type-A connector or receptacle. Alternatively, the first display panel **104** may be coupled to the main SoC **108** or the housing **109** of the main SoC **108** using any other type of cable (e.g., an HDMI mini cable, an HDMI micro cable, a universal serial bus (USB) cable (e.g., a USB Type-C cable)), communication protocol, and/or communication interface. Alternatively, the display interface **110** of FIG. 1 may have any other quantity and/or type of receptacle.

[0034] The main SoC **108** of the example of FIG. 1 is host hardware. For example, the main SoC **108** is a host (e.g., a computing host) that includes hardware to generate video data for presentation on the display panel **104**. The main SoC **108** may be implemented by one or more integrated circuits (ICs) (e.g., compact ICs) that incorporate components of a computer and/or other electronic system in a compact format. For example, the main SoC **108** may be implemented by a

combination of one or more program processors, hardware logic, and/or hardware peripherals and/or interfaces. The main SoC **108** of the example of FIG. **1** includes one or more example central processing units (CPUs) **114**, one or more example graphic processing units (GPUs) **116**, an example power controller **118**, and an example video display controller (VDC) **120**. The CPU(s) **114**, the GPU(s) **116**, the power controller **118**, and the VDC **120** are communicatively coupled to one or more example bus(es) **122**. For example, the bus(es) **122** may be implemented by at least one of an I2C bus, a SPI bus, a Peripheral Component Interconnect (PCI) bus, or any other communication protocol between integrated circuits (ICs). In this example, the power controller **118** and/or the VDC **120** may be implemented by hardware, software, and/or firmware.

[0035] Additionally or alternatively, the example main SoC **108** of FIG. **1** may include memory (e.g., flash memory, random access memory (RAM), etc.), input/output (I/O) port(s), and/or secondary storage. For example, the main SoC **108** includes the one or more CPUs **114**, the one or more GPUs **116**, the power controller **118**, the VDC **120**, the bus(es) **122**, the memory, the I/O port(s) and device(s), and/or the secondary storage all on the same substrate. In some examples, the main SoC **108** includes digital, analog, mixed-signal, radio frequency (RF), or other signal processing functions. In some examples, at least one of the one or more CPUs **114**, the one or more GPUs **116**, the power controller **118**, the VDC **120**, the bus(es) **122**, the memory, the I/O port(s), or the secondary storage may be virtualized, such as by being implemented by a virtual machine, a hyper-converged infrastructure, etc., and/or a combination thereof.

[0036] The VDC **120** may be implemented by one or more integrated circuits. The VDC **120** of the example of FIG. **1** implements a video signal generator, which is responsible for the generation of a video signal, and, in some examples, an audio signal. In this example, the host OS of the main SoC **108** controls the VDC **120** with one or more drivers (e.g., device drivers) executed by the OS. In some examples, the VDC **120** generates video signals by generating the timing of the video signals, such as the horizontal and vertical synchronization signals and the blanking interval signal. In some examples, the VDC **120** generates video signals in accordance with a digital display, an analog display (e.g., a CRT display), etc. In some examples, the GPU(s) **116** generates pixel data (e.g., pixels) for assembly and/or compilation into video signals by the VDC **120**. In such examples, the VDC **120** transmits the video signals as video data to the first display panel **104** for rendering and/or otherwise presentation to a user, such as the human **130**.

[0037] In the example of FIG. **1**, the first display panel **104** includes the first human detection system **102**. The first human detection system **102** includes an example human presence sensor (HPS) **124**, an example display SoC (e.g., a display panel SoC, a display device SoC, etc.) **126**, and example human detection logic (HDL) **128**. Alternatively, the first display panel **104** may not include the HPS **124**, the display SoC **126**, and/or the HDL **128**. For example, one or more of the HPS **124**, the display SoC **126**, and/or the HDL **128** may be external to the first display panel **104**.

[0038] The HPS **124** of this example is a proximity sensor, such as an infrared sensor. For example, the HPS **124** adjusts an output electrical signal (e.g., a voltage, a current, etc.) by detecting a change in inductance, capacitance, photoelectric field, magnetic field, etc., and/or a combination thereof, in response to a presence or absence of an object (e.g., the human **130**) proximate and/or otherwise near the HPS **124**. Alternatively, the HPS **124** may be a time-of-flight laser sensor, an ultrasonic sensor, a millimeter wave radar sensor, or a camera or camera-based sensor.

[0039] In some examples, the HPS **124** implements means for generating (e.g., first means for generating) a detection signal based on whether a human is detected. In such examples, the first means for generating may be implemented by one or more analog or digital circuit(s), logic circuits, programmable processor(s), programmable controller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)) and/or field programmable logic device(s) (FPLD(s)), proximity sensor(s), time-of-flight laser sensor(s), ultrasonic sensor(s), millimeter wave radar sensor(s), or camera(s) or camera-based sensor(s).

[0040] The display SoC **126** is display device hardware coupled to an example housing **125** of the first display panel **104** and/or to a base or stand coupled to the housing **125** and/or otherwise associated with the first display panel **104**. The display SoC **126** may be implemented by one or more programmable processors, hardware logic, one or more hardware peripherals, one or more hardware interfaces, etc., and/or a combination thereof. For example, the display SoC **126** implements a display OS representative of software and/or firmware executing on display hardware to control the first display panel **104**.

[0041] A first input (e.g., an example HPS signal **132**) of the HDL **128** is communicatively coupled to and/or in circuit with an output of the HPS **124**. A second input (e.g., an example hot plug detect (HPD) signal **140**) of the HDL **128** is coupled to and/or otherwise in circuit with an output of the display SoC **126**. An output of the HDL **128** is communicatively coupled to and/or in circuit with the receptacle **112**. The HDL **128** of this example is logic circuitry. For example, the HDL **128** is implemented with hardware, which may be represented by software and/or firmware. In such examples, the HDL **128** may be implemented by one or more analog or digital circuit(s), logic circuits and gates, programmable processor(s), programmable controller(s), graphics processing unit(s) (GPU(s)), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)) and/or field programmable logic device(s) (FPLD(s)).

[0042] In the illustrated example of FIG. **1**, the HDL **128** is coupled to a pin (e.g., an electrical pin, a connector pin, etc.) of the receptacle **112**. For example, the HDL **128** is coupled to the VDC **120**, and/or, more generally, the main SoC **108**, via a pin of the receptacle **112**. For example, the display SoC **126** is coupled to the VDC **120**, and/or, more generally, to the main SoC **108**, via one or more pins of the receptacle **112**. In such examples, the display SoC **126** transmits and/or receives example display data **127**, such as Digital Visual Interface (DVI) signals, HDMI signals, etc.

[0043] In some examples, the display data **127** includes video data (e.g., pixel data) to be presented on the first display panel **104**. In some examples, the display data **127** includes communication signals (e.g., auxiliary (AUX) signals) or data that are transmitted over communication channel(s) (e.g., AUX channel(s)). In such examples, the one or more different pins of the receptacle **112** are AUX pins or ports to facilitate communication between the main SoC **108** and the display SoC **126** over one or more AUX channels. For example, the host OS of the main SoC **108** is in communication with the display SoC **126** (e.g., the display OS of the display SoC **126**) via the AUX pin(s) of the receptacle **112**, the AUX channel(s) of the display interface **110**, etc.

[0044] The first computing environment **100** includes the first human detection system **102** to adjust a power state of the first display panel **104** based on whether one or more humans (e.g., users, human users, etc.) **130** are absent or present. For example, the first human detection system **102** causes the first display panel **104** to adjust from a first power state to a second power state in response to detecting an absence of the human **130** and/or in response to not detecting a human. In such examples, the first power state occurs when the first display panel **104** is turned on and, thus, in a powered or enabled state, and the second power state occurs when the first display panel **104** is turned off and, thus, in a powered off, reduced power, sleep, or disabled state. In other examples, the first power state is a first brightness level, a first backlight intensity value, etc., and the second power state is a second brightness level, a second backlight intensity value, etc. In some such examples, the second brightness level, the second backlight intensity value, etc., is less than the first brightness level, the first backlight intensity value, etc.

[0045] In some examples, the first human detection system **102** causes the first display panel **104** to adjust from the second power state to the first power state in response to detecting the human **130**. For example, the first human detection system **102** of the illustrated example causes the first display panel **104** to turn on in response to detecting the human **130**. In other examples, the first human detection system **102** causes the first display panel **104** to increase a brightness level, a backlight intensity value, etc., in response to detecting the human **130**.

[0046] The first human detection system **102** includes the HPS **124** to detect and/or otherwise determine whether the human **130** is present and/or otherwise proximate the first display panel **104**. The HPS **124** generates an example HPS signal **132** and an example interrupt request (IRQ) signal **134** based on the detection. The HPS signal **132** is a detection signal indicative of whether the HPS **124** detects a presence of the human **130**. For example, the HPS **124** of the illustrated example transmits the HPS signal **132** to the HDL **128** and the IRQ signal **134** to the display SoC **126**. In some such examples, the display SoC **126** determines whether the human **130** is detected based on the IRQ signal **134**.

[0047] In some examples, the IRQ signal **134** is a signal that is toggled (e.g., from a logic low signal to a logic high signal, from a logic high signal to a logic low signal, etc.) based on a state change of the HPS **124**. For example, the HPS **124** asserts the IRQ signal **134** in response to a change from detecting a presence of the human **130** to detecting an absence of the human **130**, a change from detecting an absence of the human **130** to detecting a presence of the human **130**, etc. In such examples, in response to an assertion of the IRQ signal **134** based on the change, the display SoC **126** obtains the HPS data **138**, which includes the HPS signal **132**, from the HPS **124**. In some such examples, the display SoC **126** instructs the HPS **124** to reset the IRQ signal **134** from a logic high signal to a logic low signal in response to obtaining the HPS data **138**.

[0048] In some examples, the HPS **124** generates the HPS signal **132** to have a first value (e.g., a logic high value representative of an analog or digital '1', a first voltage, a first current, etc.) in response to detecting an absence of the human **130**. The HPS **124** also asserts the IRQ signal **134** as described above in response to detecting an absence of the human **130**. These two signals are generated simultaneously or asynchronously with a time lag between each other.

[0049] In some examples, the HPS **124** generates the HPS signal **132** to have a second value (e.g., a logic low value representative of an analog or digital '0', a second voltage, a second current, etc.) in response to detecting the human **130** being proximate and/or otherwise present near the first display panel **104**, where the second value is different from the first value. The HPS **124** also asserts the IRQ signal **134** as described above in response to detecting the human **130** being proximate and/or otherwise present near the first display panel **104**. These two signals are generated simultaneously or asynchronously with a time lag between each other.

[0050] In the illustrated example of FIG. 1, the HPS **124** communicates with the display SoC **126** via an example bus (e.g., a data bus, a data interface bus, etc.) **136**. For example, the bus **136** is an I2C bus that can transport the HPS data **138** generated by the HPS **124** to the display SoC **126**. In other examples, the bus **136** can be a SPI bus or any other bus and/or communication protocol. In some examples, the HPS **124** transmits example HPS data **138** to the display SoC **126**. For example, the HPS data **138** is detection data and/or metadata that includes one or more detection parameters representative of a configuration (e.g., an HPS configuration) of the HPS signal **132**. In some examples, the detection data, the detection metadata, and/or the detection parameters (referred to herein generally as the HPS data **138**) includes a current or instant distance measurement reflecting a spacing between the HPS **124** and the human **130**, an indicator or alert representative of whether the human **130** is present or absent (e.g., a human detection alert or indicator when the human **130** is present and/or not detected as absent, a human absence alert or indicator when the human **130** is absent and/or not detected as present, etc.), a presence confidence value (e.g., a quantification of how confident the HPS **124** is that the human **130** is either absent or present), an attention level (e.g., how attentive the human **130** is with respect to the first display panel **104**), an attention confidence value (e.g., a quantification of how confident the HPS **124** is that the human **130** has a specified level of attentiveness with respect to the first display panel **104**), a threshold of detection (e.g., a threshold distance of the human **130**, a threshold presence confidence, etc., and/or a combination thereof), HPS capabilities (e.g., whether the HPS **124** supports determinations or measurements of distance, presence, and/or attentiveness, etc.), etc., and/or a combination thereof.

[0051] In the illustrated example of FIG. 1, the display SoC **126** is in communication with and/or coupled to the HDL **128**. In this example, the display SoC **126** generates the example HPD signal (e.g., a panel HPD signal) **140**. The HPD signal **140** of this example is a connection status signal representative of and/or otherwise corresponding to whether the first display panel **104** and the main SoC **108** are in communication with and/or otherwise connected to each other. Additionally or alternatively, the HPD signal **140** of this example may be a power connection signal representative of and/or otherwise corresponding to whether the first display panel **104** and the main SoC **108** are powered and connected to each other. For example, the HPD signal **140** is a first value (e.g., a logic one, a first voltage, a first current, etc.) in response to the display SoC **126** and the main SoC **108** being powered and connected to each other. In other examples, the HPD signal **140** is a second value (e.g., a logic zero, a second voltage, a second current, etc.) in response to the display SoC **126** and the main SoC **108** being unpowered and/or when the main SoC **108** and the display SoC **126** are disconnected.

[0052] In some examples, the HDL **128** compares the HPS signal **132** and the HPD signal **140** and generates an example HDL signal **142** based on the comparison. In such examples, the HDL **128** is implemented by at least a comparator (e.g., a comparator circuit, a comparator logic circuit, etc.) to compare the HPS signal **132** and the HPD signal **140** to generate the HDL signal **142**. The HDL signal **142** of this example is a control signal indicative of a desired or intended adjustment of a power state of a display panel, such as the first display panel **104**. For example, the HDL **128** generates a first value for the HDL signal **142** in response to determining that (1) the display SoC **126** is connected to the main SoC **108** based on the HPD signal **140** and (2) the HPS **124** detects the human **130**. In some examples, the HDL **128** generates a second value for the HDL signal **142** in response to determining that (1) the display SoC **126** is connected to the main SoC **108** based on the HPD signal **140** and (2) the HPS **124** does not detect the human **130**. In some examples, the HDL **128** generates the second value for the HDL signal **142** in response to determining that the display SoC **126** is not connected to the main SoC **108** based on the HPD signal **140**.

[0053] In some examples, the display SoC **126** adjusts a power state of the first display panel **104** based on the HPS signal **132**. For example, the display SoC **126** reduces an example panel backlight **144** in response to the human **130** not being present based on the HPS signal **132**. In other examples, the display SoC **126** increases the panel backlight **144** in response to the human **130** being present based on the HPS signal **132**. In such examples, the display SoC **126** adjusts whether to generate pulse-width modulation (PWM) signals to control the panel backlight **144**, control presentation of video data, frames, etc., on the first display panel **104**, etc.

[0054] In some examples, the display SoC **126** adjusts a power state of the first display panel **104** based on the HPS data **138**. For example, the display SoC **126** reduces the panel backlight **144** in response to the human **130** not being present based on the HPS data **138**. In other examples, the display SoC **126** increases the panel backlight **144** in response to the human **130** being present based on the HPS data **138**. In such examples, the display SoC **126** adjusts whether to generate PWM signals to control the panel backlight **144**, control presentation of video data, frames, etc., on the first display panel **104**, etc.

[0055] In some examples, the HDL **128**, and/or, more generally, the first display panel **104**, instructs and/or otherwise invokes the main SoC **108** to adjust transmission of the display data **127**. For example, the HDL **128** transmits the HDL signal **142** to the VDC **120**. In such examples, the VDC **120** causes at least one of the CPUS(s) **114** or the GPU(s) **116** to either begin generating or stop generating video data, frames, etc., based on the HDL signal **142**. For example, the HDL **128** instructs and/or otherwise invokes the VDC **120** to transmit video data to the first display panel **104** in response to the human **130** being detected. In other examples, the HDL **128** instructs and/or otherwise invokes the VDC **120** to stop or cease transmitting video data to the first display panel **104** in response to the human **130** not being detected.

[0056] In some examples, the main SoC **108** enables or disables the first human detection system

102 or portion(s) thereof. For example, the main SoC **108** transmits an enable signal to the display SoC **126** over the AUX channel. In such examples, the display SoC **126** enables at least one of the HPS **124** or the HDL **128** based on the enable signal. In other examples, the main SoC **108** transmits a disable signal to the display SoC **126** over the AUX channel. In such examples, the display SoC **126** disables at least one of the HPS **124** or the HDL **128** based on the disable signal. In some such examples, the main SoC **108** re-enables the at least one of the HPS **124** or the HDL **128**. For example, after the disabling of the HPS **124** and/or the HDL **128**, the main SoC **108** transmits an enable signal (e.g., a re-enable signal) to the display SoC **126** over the AUX channel. In such examples, the display SoC **126** re-enables at least one of the HPS **124** or the HDL **128**. [0057] Advantageously, the main SoC **108** effectuates backwards compatibility for legacy devices. For example, the main SoC **108** queries the display SoC **126** over the AUX channel for whether the first display device **104** includes the first human detection system **102** or portion(s) thereof. In such examples, in response to the first display device **104** not including the first human detection system **102** or portion(s) thereof (e.g., not including the HPS **124**, the HDL **128**, etc.) based on the query, the main SoC **108** transmits display data for presentation on the first display device **104** (e.g., not based on whether the human **130** is detected). For example, the main SoC **108** operates as described below in connection with the example of FIG. 3. In other examples, in response to the first display device **104** including the first human detection system **102** or portion(s) thereof (e.g., including the HPS **124**, the HDL **128**, etc.) based on the query, the main SoC **108** transmits display data for presentation on the first display device **104** based on whether the human **130** is detected by the first human detection system **102**.

[0058] In some examples, the main SoC **108** effectuates backwards compatibility when different display devices are connected to the main SoC **108**. For example, in response to the first display device **104** being disconnected from the main SoC **108**, the main SoC **108** maintains operation (e.g., executing a host OS without transmitting display data for presentation). In such examples, in response to a second display device being connected to the main SoC **108** that does not include the first human detection system **102** or portion(s) thereof, the main SoC **108** transmits display data for presentation on the second display device (e.g., not based on whether the human **130** is detected). For example, the main SoC **108** operates as described below in connection with the example of FIG. 3.

[0059] FIG. 2 is a schematic illustration of a second example computing environment **200** including a second example human detection system **202** included in a second example display panel **204**. The second human detection system **202** includes the HPS **124** and the display SoC **126** of FIG. 1. Alternatively, the second display panel **204** may not include the HPS **124** and/or the display SoC **126**. For example, the HPS **124** and/or the display SoC **126** may be external to the second display panel **204**.

[0060] The second computing environment **200** includes a second example computing system **206**, which includes the second display panel **204** and the main SoC **108** of FIG. 1. In the example of FIG. 2, the second display panel **204** is coupled to the main SoC **108** via the receptacle **112** and the display interface **110** of FIG. 1.

[0061] The second display panel **204** includes the second human detection system **202** to detect whether the human **130** of FIG. 1 is present or absent. The HPS **124** is coupled to the display SoC **126**. In this example, the HPS **124** transmits the HPS signal **132** of FIG. 1. In this example, the HPS **124** transmits the HPS data **138** to the display SoC **126** via the bus **136**.

[0062] In some examples, the HPS **124** generates the HPS signal **132** to be indicative of human detection in response to detecting the human **130**. In such examples, the HPS **124** transmits the HPS signal **132** to the display SoC **126**. In some such examples, the display SoC **126** queries the HPS **124** for the HPS data **138** and, in response to the query, the HPS **124** transmits the HPS data **138** to the display SoC **126**. In some examples, the HPS **124** transmits the display data **127** to the main SoC **108** with the help of display SoC **126**. For example, in response to the display SoC **126**

receiving the HPS data **138**, the display SoC **126** transmits the display data **127** to the main SoC **108**, where the display data **127** includes one or more portions of the HPS data **138**. In some examples, the display SoC **126** of the second human detection system **202** generates the display data **127** to include at least one of (1) first data based on and/or otherwise including the HPS signal **132** or (2) second data based on and/or otherwise including the HPS data **138**. For example, the display SoC **126** transmits the display data **127** over the AUX channel of the display interface **110**. [0063] In some examples, the main SoC **108** determines whether to adjust a power state of the display panel **204** based on the display data **127**, which includes the HPS data **138**. For example, the power controller **118** of the main SoC **108** generates, maintains, and/or updates power profile(s) (e.g., power management profile(s)) associated with at least one of the first display panel **104** of FIG. 1, the second display panel **204** of FIG. 1, and/or portion(s) thereof, such as the HPS **124**, the display SoC **126**, etc. In such examples, the power profile includes one or more policies, configurations, settings, etc., to effectuate power management of at least one of the first display panel **104** of FIG. 1, the second display panel **204** of FIG. 1, or the portion(s) thereof. For example, the power profile includes power management settings for when to turn on or off the second display panel **204**, adjust a brightness level of the second display panel **204**, adjust a backlight intensity of the second display panel **204**, etc., based on at least the HPS data **138**.

[0064] In some examples, the power controller **118**, and/or, more generally, the main SoC **108**, determines to adjust a power state of the second display panel **204** based on the HPS data **138**. For example, in response to the power controller **118** determining that the human **130** has been detected based on the HPS data **138**, the power controller **118** invokes at least one of the CPU(s) **114** or the GPU(s) **116** to generate video data to be rendered and/or otherwise presented on the second display panel **204**. In such examples, the VDC **120** transmits the video data to the display SoC **126** as the display data **127**. In other examples, in response to the power controller **118** determining that the human **130** has not been detected based on the HPS data **138**, the power controller **118** invokes at least one of the CPU(s) **114** or the GPU(s) **116** to stop generating video data. In such examples, the VDC **120** stops transmitting the video data to the display SoC **126** as the display data **127**.

[0065] In some examples, the main SoC **108** enables or disables the second human detection system **202** or portion(s) thereof. For example, the main SoC **108** transmits an enable signal to the display SoC **126** over the AUX channel. In such examples, the display SoC **126** enables the HPS **124** based on the enable signal. In other examples, the main SoC **108** transmits a disable signal to the display SoC **126** over the AUX channel. In such examples, the display SoC **126** disables the HPS **124** based on the disable signal. In some such examples, the main SoC **108** re-enables the HPS **124**. For example, after the disabling of the HPS **124**, the main SoC **108** transmits an enable signal (e.g., a re-enable signal) to the display SoC **126** over the AUX channel. In such examples, the display SoC **126** re-enables the HPS **124**.

[0066] Advantageously, the main SoC **108** effectuates backwards compatibility for legacy devices. For example, the main SoC **108** queries the display SoC **126** over the AUX channel for whether the second display device **204** includes the second human detection system **202** or portion(s) thereof. In such examples, in response to the second display device **204** not including the second human detection system **202** or portion(s) thereof (e.g., not including the HPS **124**) based on the query, the main SoC **108** transmits display data for presentation on the second display device **204** (e.g., not based on whether the human **130** is detected). For example, the main SoC **108** operates as described below in connection with the example of FIG. 3. In other examples, in response to the second display device **204** including the second human detection system **202** or portion(s) thereof (e.g., including the HPS **124**) based on the query, the main SoC **108** transmits display data for presentation on the second display device **204** based on whether the human **130** is detected by second human detection system **202**.

[0067] In some examples, the main SoC **108** effectuates backwards compatibility when different display devices are connected to the main SoC **108**. For example, in response to the second display

device **204** being disconnected from the main SoC **108**, the main SoC **108** maintains operation (e.g., executing a host OS without transmitting display data for presentation). In such examples, in response to a third display device being connected to the main SoC **108** that does not include the second human detection system **202** or portion(s) thereof, the main SoC **108** transmits display data for presentation on the third display device (e.g., not based on whether the human **130** is detected). For example, the main SoC **108** operates as described below in connection with the example of FIG. **3**.

[0068] FIG. **3** is a block diagram of a third computing system **300** including a third display panel **302**, a SoC **304**, and an HPS **306**. The third display panel **302** is coupled to the SoC **304** with a display interface **308**, such as an Embedded DisplayPort (eDP) interface, an HDMI, etc. The HPS **306** is coupled to the SoC **304** with an HPS interface **310**. In some examples, the HPS interface **310** may implement one or more interfaces, such as an I2C interface, a SPI interface, and/or a GPIO interface. Additionally or alternatively, the HPS interface **310** may implement a Universal Asynchronous Receiver/Transmitter (UART) interface. In this example, the third display panel **302** is an LED display, an OLED display, an LCD, a CRT display, an IPS display, a touchscreen, etc. In this example, the HPS **306** is an infrared sensor, a time-of-flight laser sensor, an ultrasonic sensor, a millimeter wave radar sensor, or a camera or camera-based sensor. In some examples, the HPS **306** corresponds to the HPS **124** of FIGS. **1** and/or **2**.

[0069] The SoC **304** is host hardware. In this example, the SoC **304** corresponds to and/or otherwise be representative of one or more programmable processors, hardware logic, one or more hardware peripherals, one or more hardware interfaces, etc., and/or a combination thereof. The SoC **304** includes CPU(s) **312**, GPU(s) **314**, a VDC **316**, a power controller **318**, and an integrated sensor hub (ISH) **320**. For example, the third display panel **302** is coupled to the VDC **316** via the display interface **308**. In other examples, the HPS **306** is be coupled to the ISH **320** via the HPS interface **310**. At least one of the CPU(s) **312**, the GPU(s) **314**, the VDC **316**, the power controller **318**, or the ISH **320** is coupled to each other via a bus **322**. For example, the bus **322** corresponds to at least one of an I2C bus, a SPI bus, or a PCI bus.

[0070] In some examples, the CPU(s) **312** correspond(s) to the CPU(s) **114** of FIGS. **1** and/or **2**. In some examples, the GPU(s) **314** correspond(s) to the GPU(s) **116** of FIGS. **1** and/or **2**. In some examples, the VDC **316** corresponds to the VDC **120** of FIGS. **1** and/or **2**. In some examples, the power controller **318** corresponds to the power controller **118** of FIGS. **1** and/or **2**.

[0071] The HPS **306** is a peripheral device that executes with its own driver and CPU cycles. In some examples, the HPS **306** detects human absence or presence. In some examples, in response to the detection of the human absence or presence, the HPS **306** (and the HPS driver) triggers the host OS of the SoC **304**. In some examples, in response to triggering the host OS, the host OS triggers the third display panel **302** to turn on or off via the display interface **308**.

[0072] The ISH **320** is hardware, firmware, and/or software that manages the operation lifecycle of the HPS **306** and interprets the data from the HPS **306**. In some examples, the ISH **320** transfers the data from the HPS **306** to the middleware library or hardware abstraction layer (HAL) of the SoC **304** via a driver (e.g., a device driver, a hardware driver, etc.) running and/or otherwise executing in the OS.

[0073] Advantageously, the first human detection system **102** is an improvement over the third computing system **300** of FIG. **3** because the HPS **124** is included in the first display panel **104** and the display SoC **126** does not have to invoke the host OS of the main SoC **108** to read data from the HPS **124** and inform the display SoC **126** to adjust a power state of the first display panel **104**. Advantageously, the second human detection system **202** is an improvement over the third computing system **300** of FIG. **3** because the HPS signal **132** is integrated into the HPD signal **140** to inform the main SoC **108** to adjust a power state of the second display panel **204**. In such examples, the second human detection system **202** is an improvement because the second display panel **204** includes the HPS **124** while the third display panel **302** may not include the HPS **306**.

because the third display panel **302** does not have I2C, GPIO, etc., connections to the ISH **320** of the SoC **304** and, thus, the third display panel **302** may not support the HPS **306** being included in the third display panel **302**.

[0074] FIG. **4** is a block diagram of an example implementation of a third example human detection system **400** including a human detection controller **405** to facilitate power management of the first example display panel **104** of FIG. **1** and/or the second example display panel **204** of FIG. **2**. The third human detection system **400** of FIG. **4** implements the first human detection system **102** of FIG. **1**, the second human detection system **202** of FIG. **2**, and/or portion(s) thereof, such as the display SoC **126**, the HDL **128**, etc., of FIGS. **1** and/or **2**. For example, the display SoC **126** of FIGS. **1** and/or **2** may implement the third human detection system **400** or portion(s) thereof.

[0075] The third human detection system **400** of the example of FIG. **4** includes the human detection controller **405** to detect an absence or presence of a user, such as the human **130** of FIGS. **1** and/or **2**, and adjust a power state of a display device, such as the first display panel **104** and/or the second display panel **204**, based on the detection. Advantageously, in some examples, the human detection controller **405** reduces a consumption of power, an execution and/or utilization of hardware, software, and/or firmware resources, etc., of the display device when a user is not detected by the human detection controller **405**.

[0076] The human detection controller **405** of the example of FIG. **4** includes an example sensor configurator **410**, an example sensor interface **420**, an example human detector **430**, an example power profiler **440**, an example command generator **450**, an example alert generator **460**, and an example database **470**, which includes example sensor configuration(s) **472** and example power profile(s) **474**. The human detection controller **405** of the example of FIG. **4** is in communication with an example human presence sensor **480** and an example SoC **490**. For example, the human detection controller **405** of the example of FIG. **4** may be implemented by the processor platform **1500** of the example of FIG. **15**.

[0077] In some examples, the human presence sensor **480** of FIG. **4** implements the HPS **124** of FIGS. **1** and/or **2**. In some examples, the HPS **480** includes and/or otherwise implements the human detection controller **405** or portion(s) thereof. For example, the HPS **480** may include and/or otherwise implement one or more of the sensor configurator **410**, the sensor interface **420**, the human detector **430**, the power profiler **440**, the command generator **450**, the alert generator **460**, the database **470**, the sensor configuration(s) **472**, and/or the power profile(s) **474**. For example, the human detection controller **405** included in the human presence sensor **480** of the example of FIG. **4** may be implemented by the processor platform **1600** of the example of FIG. **16**.

[0078] In some examples, the SoC **490** implements the display SoC **126** of FIGS. **1** and/or **2**. In some examples, the SoC **490** includes and/or otherwise implements the human detection controller **405** or portion(s) thereof. For example, the SoC **490** may include and/or otherwise implement one or more of the sensor configurator **410**, the sensor interface **420**, the human detector **430**, the power profiler **440**, the command generator **450**, the alert generator **460**, the database **470**, the sensor configuration(s) **472**, and/or the power profile(s) **474**. For example, the human detection controller **405** of included in the SoC **490** may be implemented by the processor platform **1700** of the example of FIG. **17**.

[0079] The human detection controller **405** of the example of FIG. **4** includes the sensor configurator **410** to initialize (e.g., power initialize), program, modify, and/or otherwise adjust a configuration of the HPS **124** of FIGS. **1** and/or **2**. In some examples, the sensor configurator **410** initializes the HPS **124** by invoking the HPS **124** to turn on and begin detection (e.g., begin detection either periodically or asynchronously). In some examples, the sensor configurator **410** transmits a command to the HPS **124** to turn off or stop human presence detection (e.g., when there is an input from an input device, such as a keyboard or mouse input). In some examples, the sensor configurator **410** transmits a command to the HPS **124** to turn on or begin human presence detection (e.g., when there is no keyboard or mouse input).

[0080] In some examples, the sensor configurator **410** adjusts one or more parameters (e.g., detection parameters), settings (e.g., detection settings), etc., of the HPS **124** to generate the sensor configuration(s) **472**. For example, the sensor configurator **410** configures the HPS **124** based on detection parameters. In such examples, in response to programming the HPS **124** with the sensor configuration(s) **472**, the sensor configurator **410** stores the sensor configuration(s) **472** in the database **470**. For example, the sensor configurator **410** instructs the HPS **124** to adjust an interval (e.g., a detection interval) when the HPS **124** detects for human presence. In other examples, the sensor configurator **410** programs the HPS **124** to detect human presence at a specific range (e.g., a range of distances), detect a human versus an object (e.g., program the HPS **124** to determine that fixed objects are not real humans), etc. In yet other examples, the sensor configurator **410** programs the HPS **124** to detect an attention level of the human **130**.

[0081] In some examples, the sensor configurator **410** facilitates an enabling or disabling of the third human detection system **400** or portion(s) thereof. For example, the SoC **490** transmits an enable signal to the human detection controller **405**. In such examples, the sensor configurator **410** enables the HPS **480** and/or other component(s) based on the enable signal. In other examples, the SoC **490** transmits a disable signal to the human detection controller **405**. In such examples, the sensor configurator **410** disables the HPS **480** and/or other component(s) based on the disable signal. In some such examples, the SoC **490** re-enables the HPS **480** and/or the other component(s). For example, after the disabling of the HPS **480** and/or the other component(s), the SoC **490** transmits an enable signal (e.g., a re-enable signal) to the human detection controller **405**. In such examples, the sensor configurator **410** re-enables the HPS **124** and/or the other component(s).

[0082] In some examples, the sensor configurator **410** implements means for configuring the HPS **124** based on detection parameters. In such examples, the means for configuring is implemented by any processor structured to perform the corresponding operation by executing software or firmware, or hardware circuit (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, a PLD, a FPLD, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

[0083] The human detection controller **405** of the example of FIG. **4** includes the sensor interface **420** to receive and/or otherwise obtain data or measurements (e.g., detection data, human presence data, human absence data, human detection data, etc.) from the HPS **124**. In some examples, the sensor interface **420** obtains at least one of a distance of the human **130**, an indicator of whether the human **130** is present or absent, an indicator of whether the human **130** is attentive, a quantification of how attentive the human **130** is to the presentation of video data on the first display panel **104**, the second display panel **204**, etc., or a capability of the HPS **124** (e.g., whether the HPS **124** supports distance, presence, and/or attention measurements or determinations).

[0084] The human detection controller **405** of the example of FIG. **4** includes the human detector **430** to determine whether an absence or presence of a user, such as the human **130**, is detected based on a measurement, detection data, etc., from the HPS **480**. In some examples, the human detector **430** indicates whether the human **130** is present based on an indicator (e.g., a detection indicator), a raw signal or raw data, etc., from the HPS **480**, etc. In such examples, the human detector **430** processes the indicator, the raw signal or the raw data, etc., by executing a noise-reduction filter, removing or reducing signal error, etc. In some such examples, the human detector **430** generates a logic high signal (e.g., a voltage representative of a digital '1'), a logic low signal (e.g., a voltage representative of a digital '0'), etc., as the HPS signal **132** of FIGS. **1** and/or **2**. In some examples, the human detector **430** determines whether the human **130** is present based on (1) the detection indicator and (2) a presence confidence value from the HPS **124**.

[0085] In some examples, the human detector **430** starts and/or otherwise triggers a timer (e.g., a change state timer, a change power state timer, etc.) that, when elapsed, causes an adjustment in a power state of the display panel **104**, **204** in response to detecting human absence. For example, the

sensor interface **420** obtains an alert indicative of the human **130** not being detected. In such examples, instead of changing the power state of the display panel **104, 204** when absence of the human **130** is detected, the human detector **430** improves a user experience associated with the display panel **104, 204** by waiting until a pre-determined time period (e.g., the timer) has elapsed before turning off the display panel **104, 204**. In some such examples, the human detector **430** reduces the likelihood of repeatedly toggling the display panel **104, 204** between an on and off power state to avoid a poor or bad user experience.

[0086] In some examples, the human detector **430** implements means for triggering a timer. In such examples, the means for triggering is implemented by any processor structured to perform the corresponding operation by executing software or firmware, or hardware circuit (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, a PLD, a FPLD, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

[0087] In some examples, the human detector **430** determines whether the human **130** is attentive to the first display panel **104**, the second display panel **204**, etc., and, if so, a quantification or representation of how attentive the human **130** is to the first display panel **104**, the second display panel **204**, etc. For example, the HPS **124** transmits attention data (e.g., human attention data) including an attention level value (e.g., a value in a range of 0-2 with 0 being representative of the human **130** being fully attentive and 2 being representative of the human **130** being less than fully attentive, a value in a range of 0-100% with 0% being representative of the human **130** not being attentive and 100% being representative of the human **130** being fully attentive, etc.). In such examples, the HPS **124** transmits an attention level value of 0 when the HPS **124** detects that the human **130** is looking at the display panel **104, 204**, an attention level value of 1 when the HPS **124** detects that a face of the human **130** is in a range of 45-90 degrees to the display panel **104, 204**, an attention level value of 2 when the HPS **124** detects that the face is greater than 90 degrees to the display panel **104, 204**, etc. In some such examples, the transmissions from the HPS **124** are implemented as the HPS data **138**. For example, the HPS data **138** may implement and/or otherwise include the attention level.

[0088] The human detection controller **405** of the example of FIG. **4** includes the power profiler **440** to generate and/or adjust the power profile(s) **474**. The power profile(s) **474** include, correspond to, and/or otherwise representative of one or more power management profiles of at least one of the first display panel **104**, the second display panel **204**, and/or portion(s) thereof, such as the HPS **124**, the display SoC **126**, etc. In some examples, the power profiler **440** generates the power profile(s) **474** to determine when to adjust a power state of the display panel **104, 204**, such as determining when to turn on or off the display panel **104, 204**, adjust a brightness or backlight intensity of the display panel **104, 204**, etc. For example, the power profiler **440** generates the power profile(s) **474** that specify triggering a timer that, when elapsed, causes an adjustment in a power state of the display panel **104, 204** in response to detecting human absence. In other examples, the power profiler **440** generates the power profile(s) **474** that specify turning on the display panel **104, 204** in response to detecting human presence, turning off the display panel **104, 204** in response to detecting human absence, etc.

[0089] The human detection controller **405** of the example of FIG. **4** includes the command generator **450** to generate a command, a direction, an instruction, an invocation, etc., to adjust a power state of the first display panel **104** and/or the second display panel **204**. In some examples, the command generator **450** generates a command to turn on or off the display panel **104, 204**. For example, the command generator **450** generates a command to invoke the display SoC **126** to stop generating the PWM signals to control the panel backlight **144** of FIGS. **1** and/or **2** and turn off power to the display panel **104, 204**. In other examples, the command generator **450** generates an instruction to direct the display SoC **126** to adjust a brightness, a backlight intensity, etc., of the display panel **104, 204**. In yet other examples, the command generator **450** transmits a command

(e.g., a de-assertion of the HDL signal **142**) to the main SoC **108** of FIG. **2** to invoke the VDC **120** of FIGS. **1** and/or **2** to stop and/or otherwise cease transmitting the display data **127** to the display SoC **126**.

[0090] In some examples, the command generator **450** implements means for generating (e.g., second means for generating) a control signal to cause an adjustment of a power state of the display device **104, 204** based on the detection signal (e.g., the HPS signal **132**). In some examples, the second means for generating is to generate the control signal to reduce a backlight intensity of the display device **104, 204** or turn off the display device **104, 204**. In some examples, the second means for generating is to determine that the detection signal is indicative of a presence of the human **130** proximate the display device **104, 204** or indicative of an absence of the human **130** proximate the display device **104, 204**, and determine that a connection status signal (e.g., the HPD signal **140**) indicates the display device **104, 204** is in communication with host hardware (e.g., the main SoC **108**). In some examples, the command generator **450** implements means for turning off the display device **104, 204** in response to a timer elapsing (e.g., a change state timer, a change power state timer, etc.). In such examples, the means for turning is to turn off the display device **104, 204** in response to the timer elapsing without the detection signal changing indication of an absence of the human **130**. In some such examples, the second means for generating and/or the means for turning off is/are implemented by any processor structured to perform the corresponding operation by executing software or firmware, or hardware circuit (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, a PLD, a FPLD, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

[0091] The human detection controller **405** of the example of FIG. **4** includes the alert generator **460** to generate an alert, an indicator, etc., representative of whether the human **130** is present or absent. For example, the alert, the indicator, etc., implements the HPS signal **132**, the IRQ signal **134**, the HPD signal **140**, and/or the HDL signal **142** of FIGS. **1** and/or **2**. In some examples, the alert generator **460** generates an alert in response to whether the human **130** is detected. For example, the alert generator **460** generates the HPS signal **132** to have a first value indicative of human presence, a second value indicative of human absence, etc. In such examples, the alert generator **460** generates the IRQ signal **134**. In some examples, the alert generator **460** generates an alert in response to detecting whether the display panel **104, 204** is powered and connected to the main SoC **108**. For example, the alert generator **460** generates the HPD signal **140** to be indicative of (1) being powered and connected to the main SoC **108** or (2) not being powered and/or not connected to the main SoC **108**. In some examples, the alert generator **460** generates an alert based on (1) whether the human **130** is detected and (2) whether the display panel **104, 204** is powered and connected to the main SoC **108**. For example, the alert generator **460** generates the HDL signal **142** based on at least one of the HPS signal **132** or the HPD signal **140**.

[0092] In some examples, the alert generator **460** implements means for generating (e.g., first means for generating) a detection signal based on whether a human is detected. In such examples, the first means for generating is implemented by any processor structured to perform the corresponding operation by executing software or firmware, or hardware circuit (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, a PLD, a FPLD, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

[0093] In some examples, the alert generator **460** implements means for asserting a connection status signal (e.g., the HPD signal **140**) in response to detecting that the display device **104, 204** is in communication with host hardware (e.g., the main SoC **108**). In some examples, the alert generator **460** implements means for de-asserting the connection status signal in response to detecting that the display device **104, 204** is in communication with the host hardware. In some examples, the alert generator **460** implements means for generating (e.g., third means for

generating) the connection status signal in response to detecting that the display device **104, 204** is in communication with the host hardware. In such examples, the means for asserting, the means for de-asserting, and/or the third means for generating is/are implemented by any processor structured to perform the corresponding operation by executing software or firmware, or hardware circuit (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, a PLD, a FPLD, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

[0094] In some examples, the alert generator **460** implements means for transmitting a control signal (e.g., the HDL signal **142**) to the host hardware to cease transmission of display data (e.g., the display data **127**) from the host hardware to the display device **104, 204**. In some examples, the means for transmitting is to transmit at least one of the detection parameters or the detection signal (e.g., the HPS signal **132**) to the host hardware in communication with the display device **104, 204**. In such examples, the means for transmitting is implemented by any processor structured to perform the corresponding operation by executing software or firmware, or hardware circuit (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, a PLD, a FPLD, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware, but other structures are likewise appropriate.

[0095] The human detection controller **405** of the example of FIG. **4** includes the database **470** to record data (e.g., the sensor configuration(s) **472**, the power profile(s) **474**, etc.). In some examples, the database **470** is implemented by a volatile memory (e.g., a Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS Dynamic Random Access Memory (RDRAM), etc.) and/or a non-volatile memory (e.g., flash memory). The database **470** may additionally or alternatively be implemented by one or more double data rate (DDR) memories, such as DDR, DDR2, DDR3, DDR4, mobile DDR (mDDR), etc. The database **470** may additionally or alternatively be implemented by one or more mass storage devices such as hard disk drive(s), compact disk (CD) drive(s), digital versatile disk (DVD) drive(s), solid-state disk drive(s), etc. While in the illustrated example the database **470** is illustrated as a single database, the database **470** may be implemented by any number and/or type(s) of databases. Furthermore, the data stored in the database **470** may be in any data format such as, for example, binary data, comma delimited data, tab delimited data, structured query language (SQL) structures, etc.

[0096] Advantageously, the SoC **490** effectuates backwards compatibility for legacy devices. For example, the SoC **490** queries a display SoC, such as the display SoC **126** of FIGS. **1** and/or **2** over an AUX channel for whether a display device includes the third human detection system **400** or portion(s) thereof. In such examples, in response to the display device not including the third human detection system **400** or portion(s) thereof (e.g., not including the HPS **480**) based on the query, the SoC **490** transmits display data for presentation on the display device (e.g., not based on whether the human **130** is detected). For example, the SoC **490** operates as described above in connection with the example of FIG. **3**. In other examples, in response to the display device including the third human detection system **400** or portion(s) thereof (e.g., including the HPS **480**) based on the query, the SoC **490** transmits display data for presentation on the display device **104** based on whether the human **130** is detected by the third human detection system **400**.

[0097] In some examples, the SoC **490** effectuates backwards compatibility when different display devices are connected to the main SoC **490**. For example, in response to a display device being disconnected from the SoC **490**, the SoC **490** maintains operation (e.g., executing a host OS without transmitting display data for presentation). In such examples, in response to a second display device being connected to the SoC **490** that does not include the third human detection system **400** or portion(s) thereof, the SoC **490** transmits display data for presentation on the second

display device (e.g., not based on whether the human **130** is detected). For example, the SoC **490** operates as described above in connection with the example of FIG. **3**.

[0098] While an example manner of implementing the first human detection system **102** and/or the second human detection system **202** of FIG. **2** is illustrated in FIG. **4**, one or more of the elements, processes and/or devices illustrated in FIG. **4** may be combined, divided, re-arranged, omitted, eliminated and/or implemented in any other way. Further, the example sensor configurator **410**, the example sensor interface **420**, the example human detector **430**, the example power profiler **440**, the example command generator **450**, the example alert generator **460**, the example database **470**, the example sensor configuration(s) **472**, the example power profile(s) **474**, and/or, more generally, the example human detection controller **400**, and/or, more generally, the third example human detection system **400** of FIG. **4**, may be implemented by hardware, software, firmware and/or any combination of hardware, software and/or firmware. Thus, for example, any of the example sensor configurator **410**, the example sensor interface **420**, the example human detector **430**, the example power profiler **440**, the example command generator **450**, the example alert generator **460**, the example database **470**, the example sensor configuration(s) **472**, the example power profile(s) **474**, and/or, more generally, the example human detection controller **405**, and/or, more generally, the third example human detection system **400**, could be implemented by one or more analog or digital circuit(s), logic circuits, programmable processor(s), programmable controller(s), GPU(s), digital signal processor(s) (DSP(s)), application specific integrated circuit(s) (ASIC(s)), programmable logic device(s) (PLD(s)) and/or field programmable logic device(s) (FPLD(s)). When reading any of the apparatus or system claims of this patent to cover a purely software and/or firmware implementation, at least one of the example sensor configurator **410**, the example sensor interface **420**, the example human detector **430**, the example power profiler **440**, the example command generator **450**, the example alert generator **460**, the example database **470**, the example sensor configuration(s) **472**, the example power profile(s) **474**, and/or, more generally, the example human detection controller **400**, is/are hereby expressly defined to include a non-transitory computer readable storage device or storage disk such as a memory, a DVD, a CD, a Blu-ray disk, etc. including the software and/or firmware. Further still, the third example human detection system **400** of FIG. **4** may include one or more elements, processes and/or devices in addition to, or instead of, those illustrated in FIG. **4**, and/or may include more than one of any or all of the illustrated elements, processes and devices. As used herein, the phrase “in communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

[0099] FIG. **5A** depicts an example truth table **500** reflecting example input and output relationships for the HDL **128** of FIGS. **1**. The truth table **500** includes a first example column **510** representative of Boolean values for input A, which corresponds to the HPS signal **132** of FIG. **1**. The truth table **500** includes a second example column **520** representative of Boolean values for input B, which corresponds to the HPD signal **140** of FIG. **1**. The truth table **500** includes a third example column **530** representative of Boolean values for output Y, which corresponds to the HDL signal **142** of FIG. **1**. Thus, a “1” represents “true” and a “0” represents “false”.

[0100] In the first column **510** of the example truth table **500** of FIG. **5**, the Boolean value of “0” represents that a human, such as the human **130** of FIG. **1**, is present, while a Boolean value of “1” represents that the human is absent with respect to the first display panel **104** of FIG. **1**. In the second column **520** of the example truth table **500** of FIG. **5**, the Boolean value of “1” represents that a panel, such as the first display panel **104**, is connected to host hardware, such as the main SoC **108** of FIG. **1**, while a Boolean value of “0” represents that the panel is disconnected from the host hardware. For example, the first display panel **104** is disconnected when the first display panel **104** is physically disconnected from the main SoC **108**. In other examples, the first display panel

104 is disconnected when the first display panel **104** is in a power off mode while the main SoC **108** is powered on. In yet other examples, the first display panel **104** is disconnected when the first display panel **104** is powered on while the main SoC **108** is in a power off mode.

[0101] In some examples, the HDL signal has a value of “1” in response to the HPS having a value of “0” and the HPD signal having a value of “1.” In such examples, the HDL signal having the value of “1” is representative of (1) the HPS **124** of FIG. 1 detecting that the human **130** is present and (2) the display SoC **126** of FIG. 1 determining that the display SoC **126** is connected to the main SoC **108** via the display interface **110** of FIG. 1.

[0102] In some examples, the HDL signal has a value of “0” in response to the HPS having a value of “1” and the HPD signal having a value of “1.” In such examples, the HDL signal having the value of “0” is representative of (1) the HPS **124** of FIG. 1 detecting that the human **130** is absent and (2) the display SoC **126** of FIG. 1 determining that the display SoC **126** is connected to the main SoC **108** via the display interface **110** of FIG. 1.

[0103] In some examples, the HDL signal has a value of “0” in response to the HPS having a value of “0” and the HPD signal having a value of “0.” In such examples, the HDL signal having the value of “0” is representative of the display SoC **126** of FIG. 1 determining that the display SoC **126** is not connected to the main SoC **108** via the display interface **110** of FIG. 1. In some such examples, the detection of the human **130** does not impact the output because the display SoC **126** is not connected to the main SoC **108**.

[0104] FIG. 5B is a schematic illustration of an example control circuit **550** implementing the HDL **128** of FIG. 1, the first human detection system **102** of FIG. 1, and/or the human detection controller **405** of FIG. 4, in accordance with the logic reflected in the example truth table **500** of FIG. 5A. The control circuit **550** of the example of FIG. 5B is implemented using a logic gate corresponding to the example truth table **500** of FIG. 5A. For example, using truth table conversion methods such as generating and analyzing Karnaugh maps, performing Boolean algebra, etc., an example Boolean expression as described in Equation (1) of FIG. 5B is generated to represent the relationships described in the truth table **500** of FIG. 5A.

[0105] Alternatively, other Boolean expressions may be generated based on the truth table **500** of FIG. 5A. In the illustrated example of Equation (1), input A corresponds to a value of the HPS signal of the first column **510**, input B corresponds to a value of the HPD signal of the second column **520**, and the output Y corresponds to a value of the HDL signal of the third column **530**. In the illustrated example of FIG. 5B, the control circuit **550** is implemented by an XOR gate **560**. The XOR gate **560** of the illustrated example represents a Boolean operator that generates the output Y based on the inputs A and B. Alternatively, additional and/or different types of logic gates may be used to implement the truth table **500** of FIG. 5A.

[0106] FIG. 6 depicts an example logic table **600** reflecting example input and output relationships for example operating conditions associated with the second human detection system **202** of FIG. 2 and/or the human detection controller **405** of FIG. 4. The logic table **600** includes a first example column **610** representative of Boolean values for an HPS signal, which corresponds to the HPS signal **132** of FIG. 2. The logic table **600** includes a second example column **620** representative of Boolean values for an HPD signal, which corresponds to the HPD signal **140** of FIG. 2. The logic table **600** includes a third example column **630** representative of Boolean values for a decision made by host hardware, which corresponds to an output generated by the main SoC **108** of FIG. 2. For example, the main SoC **108** transmits the decision via an AUX channel of a display interface, such as the display interface **110** of FIG. 2. Thus, a “1” represents “true,” a “0” represents “false,” and an “X” represents “don't care.” In the don't care condition, the corresponding input has no effect on the resulting output.

[0107] In the first column **610** of the example logic table **600** of FIG. 6, the Boolean value of “0” represents that a human, such as the human **130** of FIG. 2, is present, while a Boolean value of “1” represents that the human is absent with respect to the second display panel **204** of FIG. 2. In the

second column **620** of the example logic table **600** of FIG. **6**, the Boolean value of “1” represents that a panel, such as the second display panel **204** of FIG. **2**, is connected to host hardware, such as the main SoC **108** of FIGS. **1** and/or **2**, while a Boolean value of “0” represents that the panel is disconnected from the host hardware.

[0108] In some examples, the host hardware decision has a value of “1” in response to the HPS having a value of “0” and the HPD signal having a value of “1.” In such examples, the host hardware decision having the value of “1” is representative of the main SoC **108** determination to have the second display panel **204** remain connected to the main SoC **108** based on (1) the HPS **124** of FIGS. **1** and/or **2** detecting that the human **130** is present and (2) the display SoC **126** of FIGS. **1** and/or **2** determining that the display SoC **126** is connected to the main SoC **108** via the display interface **110** of FIGS. **1** and/or **2**.

[0109] In some examples, the host hardware decision has a value of “0” in response to the HPS having a value of “0” and the HPD signal having a value of “0.” In such examples, the host hardware decision having the value of “0” is representative of the main SoC **108** determination to have the second display panel **204** remain connected to the main SoC **108** even in response to the HPS **124** of FIGS. **1** and/or **2** detecting that the human **130** is absent. In some such examples, the main SoC **108** compares at least one of the HPS signal **132** or the HPD signal **140** to a power management profile in the host OS. For example, the main SoC **108** determines to maintain the power state of the second display panel **204** to be in an enabled, powered, and/or otherwise “ON” state.

[0110] In some examples, the main SoC **108** determines the host hardware decision by executing system wide power policy management. For example, the main SoC **108** obtains sensor information from the HPS **124** as the display data **127** over an AUX channel of an interface, such as the display interface **110**. In some examples, the main SoC **108** executes a system wide power policy framework (e.g., based on the display data **127**) and manages the power of a computing system, such as the second computing system **206** of FIG. **2**, in a way the main SoC **108** wants to conserve or save power, conserve or save compute power, change a priority of tasks (e.g., computing tasks) that are running, pause or stop the generation and/or presentation of audio and/or video data, transition the second computing system **206** into a power state (e.g., an Intel® S0ix state), etc.

[0111] If the main SoC **108** determines to ignore the human absence event (e.g., the main SoC **108** determines the human absence event to not be substantial enough to put the second computing system **206** in a low power state), then the main SoC **108** communicates back to the display SoC **126** over an AUX channel regarding the host hardware decision. If the main SoC **108** determines to execute an action in response to the human absence event (e.g., the main SoC **108** determines the human absence event is substantial enough to put the second computing system **206** in a low power state), then the main SoC **108** communicates back to the display SoC **126** over an AUX channel regarding the host hardware decision. For example, the main SoC **108** sends an update to the display SoC **126** that the main SoC **108** is to stop sending video frames (e.g., any video frames) to the second display panel **204** and that the display SoC **126** can shut down the second display panel **204** or enter into a lower power state. In such examples, the display SoC **126** executes and/or otherwise performs a power optimization sequence. For example, the display SoC **126** pulls low the HPD signal **140** over the display interface **110** to notify the main SoC **108**, so that the main SoC **108** can turn off the second display panel **204** as if the second display panel **204** was physically disconnected, unplugged, etc. Accordingly, the display SoC **126** resets the HPS **124** to transition the HPS **124** from a human absence detection state to a human presence detection state.

[0112] In some examples, in response to turning off the second display panel **204** or entering a low power state, the HPS **124** detects human presence and notifies the display SoC **126**. The display SoC **126** toggles (e.g., pulls high) the HPD signal **140** on the display interface **110** to notify the main SoC **108** that the main SoC **108** can turn on the second display panel **204** as if the second

display panel **204** was physically connected, plugged in, etc. The display SoC **126** transmits the change in human presence to the main SoC **108** over the AUX channel of the display interface **110**. In some examples, the main SoC **108** queries the display SoC **126** for additional information, such as the HPS data **138**. In some examples, the main SoC **108** executes the system wide power policy management based on at least the HPS data **138** obtained from the display SoC **126** via the AUX channel. In such examples, the main SoC **108** executes the system wide power policy framework and enable systems that were put into a lower power mode, change a priority of tasks (e.g., computing tasks) that are running, begin or trigger the generation and/or presentation of audio and/or video data, transition the second computing system **206** into a power state (e.g., an Intel® S0ix state), etc. In such examples, the main SoC **108** transmits an update to the display SoC **126** via the AUX channel and establish a link with the second display panel **204** to begin or start sending display frames to the display SoC **126**. In some such examples, the display SoC **126** executes its standard operation to establish the display panel link and start displaying the display frames sent from the main SoC **108**. Accordingly, the display SoC **126** resets the HPS **124** to transition the HPS **124** from a human presence detection state to a human absence detection state.

[0113] FIG. 7 is a block diagram of example data flow **700** across an example hardware layer (e.g., a hardware stack) **702** and an example software layer (e.g., an OS layer, an OS stack, etc.) **704** associated with the first example human detection system **102** of FIG. 1, the second example human detection system **202** of FIG. 2, and/or the third example human detection system **400** of FIG. 4. In the example of FIG. 7, an example HPS integrated display panel **706**, such as the first display panel **104** of FIG. 1 and/or the second display panel **204** of FIG. 2, transmits first HPS data (e.g., an alert or indicator of human absence or detection, detection parameters, etc.) to a physical display port (e.g., a physical display hardware port) **708** via an example display cable **710**, such as an HDMI cable or an eDP cable. In some examples, the HPS integrated display panel **706** transmits the HDL signal **142** to the physical display port **708**. In some examples, the HPS integrated display panel **706** transmits the HPD signal **140** and the display data **127**, which can include the HPS signal **132**, the HPS data **138**, etc. For example, the physical display port **708** may implement the receptacle **112** of FIGS. 1 and/or 2. The hardware layer **702** of the example of FIG. 7 includes the HPS integrated display panel **706** and the physical display port **708**.

[0114] In the example of FIG. 7, the physical display port **708** transmits the first HPS data from the display panel **706** of the hardware layer **702** to an example display panel OS driver **712** of the software layer **704** via an example kernel application programming interface (API) **714**. The display panel OS driver **716** transmits the first HPS data to an example display middleware library **718** via an example system call **720**. For example, the display middleware library **718** may be implemented by the VDC **120** of FIGS. 1 and/or 2. The display middleware library **718** transmits the first HPS data to an example platform power policy framework **722** via an example API **724**. The platform power policy framework **722** transmits the first HPS data to an example user interface (UI) application **725** via an example API **726**. For example, the UI application **725** is executed and/or otherwise presented to a user, such as the human **130** of FIGS. 1 and/or 2.

[0115] In the example of FIG. 7, an example HPS **728**, such as the HPS **124** of FIGS. 1 and/or 2, transmits second HPS data (e.g., an alert or indicator of human absence or detection) to an example sensor hub **730** via example interfaces **732**, such as an I2C, SPI, and/or GPIO interface. For example, the sensor hub **730** corresponds the ISH **320** of FIG. 3. The HPS **728** of FIG. 7 is not included in the HPS integrated display panel **706**. For example, the HPS **728** is used for a different system function than determining whether to adjust a power state of the first display panel **104**. The sensor hub **730** transmits the second HPS data to an example sensor hub OS driver **733** via an example system call **734**. The sensor hub OS driver **733** transmits the second HPS data to an example middleware library **736** via an example system call **738**. The middleware library **736** transmits the second HPS data to an example sensor software development kit (SDK) **740** via an example API **742**. The sensor SDK **740** transmits the second HPS data to the UI application **725** via

an example API **744**.

[0116] In some examples, the HPS integrated display panel **706** adjusts a power state of the HPS integrated display panel **706** based on the first HPS data. In such examples, the platform power policy framework **722** determines to adjust a transmission of video data to the first display panel **104** and/or the second display panel **204** based on the first HPS data obtained via the data flow depicted in the example of FIG. 7. Advantageously, in some examples, the HPS integrated display panel **706** adjusts the power state of the first display panel **104** independently from host hardware, such as the main SoC **108** of FIGS. 1 and/or 2. For example, there is no communication in the example of FIG. 7 between the sensor subsystem **746** and the display subsystem **748** for human presence detection. Advantageously, in some examples, the HPS integrated display panel **706** adjusts the power state of the second display panel **204** in conjunction with host hardware, such as the main SoC **108** of FIGS. 1 and/or 2. For example, there is no communication in the example of FIG. 7 between the sensor subsystem **746** and the display subsystem **748** for human presence detection.

[0117] FIG. 8 is a block diagram of example data flow **800** across an example hardware layer (e.g., a hardware stack) **802** and an example software layer (e.g., an OS layer or stack) **804** associated with the third computing system **300** of FIG. 3. In the example of FIG. 8, an HPS **806**, such as the HPS **306** of FIG. 3, transmits HPS data (e.g., an alert or indicator of human absence or detection, detection parameters, etc.) to a sensor hub **808**, such as the ISH **320** of FIG. 3, via an HPS interface **810**, such as the HPS interface **310** of FIG. 3. For example, the HPS interface **810** may be implemented by an I2C interface, a SPI interface, and/or a GPIO interface.

[0118] In the example of FIG. 8, the sensor hub **808** transmits the HPS data to a sensor hub OS driver **812** via a system call **814**. The sensor hub OS driver **812** transmits the HPS data to a middleware library **816** via a system call **818**. The middleware library **816** transmits the HPS data to a sensor SDK **820** via an API **822**. The sensor SDK **820** transmits the HPS data to a UI application **824** via an API **826**.

[0119] The sensor SDK **820** determines to adjust a power state of a display panel **828** based on the HPS data. For example, the sensor SDK **820** determines to turn off the display panel **828** in response to the HPS **806** detecting that a user is not present, turn on the display panel **828** in response to the HPS **806** detecting that the user is present, etc. The sensor SDK **820** generates a display control command indicative of the determination.

[0120] The sensor SDK **820** transmits the display control command to a display middleware library **830** via an event or system call **832**. The display middleware library **830** transmits the display control command to a display panel OS driver **834** via a system call **836**. The display panel OS driver **834** transmits the display control command to a physical display port **838** via a kernel API **840**. The physical display port **838** transmits the display control command to the display panel **828** via a display cable **842**, such as the display interface **308** of FIG. 3. In response to the display panel **828** receiving the display control command, the display panel **828** adjusts a power state of the display panel **828** based on the display control command.

[0121] Advantageously, the first human detection system **102** of FIG. 1, the second human detection system **202** of FIG. 2, and/or the third human detection system **400** of FIG. 4 are improvement(s) over the third computing system **300** of FIG. 3 because a sensor subsystem **844** has to receive the HPS data to control a display subsystem **846**. Advantageously, the first human detection system **102** of FIG. 1, the second human detection system **202** of FIG. 2, and/or the third human detection system **400** of FIG. 4 adjusts a power state of the first display panel **104** of FIG. 1 and/or the second display panel **204** of FIG. 2 by including the HPS **124** in and/or otherwise integrated with the display panel **104**, **204** and adjusting the power state with hardware, software, and/or firmware of the display panel **104**, **204** for an improved user experience and reduction(s) in power consumption and/or utilized computing resources.

[0122] Data flow diagrams and/or flowcharts representative of example hardware logic, machine

readable instructions, hardware implemented state machines, and/or any combination thereof for implementing the first example human detection system **102** of FIG. **1**, the second example human detection system **202** of FIG. **2**, and/or the third human detection system **400** of FIG. **4** is/are shown in FIGS. **9-14**. The machine readable instructions may be one or more executable programs or portion(s) of an executable program for execution by a computer processor and/or processor circuitry, such as the processor **1512** shown in the example processor platform **1500** discussed below in connection with FIG. **15**, the example processor platform **1600** discussed below in connection with FIG. **16**, and/or the example processor platform discussed below in connection with FIG. **17**. The program may be embodied in software stored on a non-transitory computer readable storage medium such as a CD-ROM, a floppy disk, a hard drive, a DVD, a Blu-ray disk, or a memory associated with the processor **1512**, the processor platform **1600**, and/or the processor platform **1700**, but the entire program and/or parts thereof could alternatively be executed by a device other than the processor **1512**, the processor platform **1700**, and/or the processor platform **1700**, and/or embodied in firmware or dedicated hardware. Further, although the example program is described with reference to the data flow diagrams and/or the flowcharts illustrated in FIGS. **9-14**, many other methods of implementing the first example human detection system **102** of FIG. **1**, the second example human detection system **202** of FIG. **2**, and/or the third human detection system **400** of FIG. **4** may alternatively be used. For example, the order of execution of the blocks may be changed, and/or some of the blocks described may be changed, eliminated, or combined. Additionally or alternatively, any or all of the blocks may be implemented by one or more hardware circuits (e.g., discrete and/or integrated analog and/or digital circuitry, an FPGA, an ASIC, a comparator, an operational-amplifier (op-amp), a logic circuit, etc.) structured to perform the corresponding operation without executing software or firmware. The processor circuitry may be distributed in different network locations and/or local to one or more devices (e.g., a multi-core processor in a single machine, multiple processors distributed across a server rack, etc.).

[0123] The machine readable instructions described herein may be stored in one or more of a compressed format, an encrypted format, a fragmented format, a compiled format, an executable format, a packaged format, etc. Machine readable instructions as described herein may be stored as data or a data structure (e.g., portions of instructions, code, representations of code, etc.) that may be utilized to create, manufacture, and/or produce machine executable instructions. For example, the machine readable instructions may be fragmented and stored on one or more storage devices and/or computing devices (e.g., servers) located at the same or different locations of a network or collection of networks (e.g., in the cloud, in edge devices, etc.). The machine readable instructions may require one or more of installation, modification, adaptation, updating, combining, supplementing, configuring, decryption, decompression, unpacking, distribution, reassignment, compilation, etc. in order to make them directly readable, interpretable, and/or executable by a computing device and/or other machine. For example, the machine readable instructions may be stored in multiple parts, which are individually compressed, encrypted, and stored on separate computing devices, wherein the parts when decrypted, decompressed, and combined form a set of executable instructions that implement one or more functions that may together form a program such as that described herein.

[0124] In another example, the machine readable instructions may be stored in a state in which they may be read by processor circuitry, but require addition of a library (e.g., a dynamic link library (DLL)), an SDK, an API, etc. in order to execute the instructions on a particular computing device or other device. In another example, the machine readable instructions may need to be configured (e.g., settings stored, data input, network addresses recorded, etc.) before the machine readable instructions and/or the corresponding program(s) can be executed in whole or in part. Thus, machine readable media, as used herein, may include machine readable instructions and/or program(s) regardless of the particular format or state of the machine readable instructions and/or program(s) when stored or otherwise at rest or in transit.

[0125] The machine readable instructions described herein can be represented by any past, present, or future instruction language, scripting language, programming language, etc. For example, the machine readable instructions may be represented using any of the following languages: C, C++, Java, C #, Perl, Python, JavaScript, HyperText Markup Language (HTML), Structured Query Language (SQL), Swift, etc.

[0126] As mentioned above, the example processes of FIGS. **9-14** may be implemented using executable instructions (e.g., computer and/or machine readable instructions) stored on a non-transitory computer and/or machine readable medium such as a hard disk drive, a flash memory, a read-only memory, a compact disk, a digital versatile disk, a cache, a random-access memory and/or any other storage device or storage disk in which information is stored for any duration (e.g., for extended time periods, permanently, for brief instances, for temporarily buffering, and/or for caching of the information). As used herein, the term non-transitory computer readable medium is expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals and to exclude transmission media.

[0127] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc. may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, and (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities and/or steps, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, and (3) at least one A and at least one B.

[0128] As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” entity, as used herein, refers to one or more of that entity. The terms “a” (or “an”), “one or more”, and “at least one” can be used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements or method actions may be implemented by, e.g., a single unit or processor. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or advantageous.

[0129] FIG. **9** is a first example data flow diagram **900** associated with the first example human detection system **102** of FIG. **1**, and/or, more generally, the first example computing environment **100** of FIG. **1**. The first data flow diagram **900** is representative of example hardware logic, machine readable instructions, hardware implemented state machines, and/or any combination thereof for implementing an example HPS **902**, an example display SoC **904**, and an example panel backlight **906**. In some examples, the HPS **902** of FIG. **9** corresponds to the HPS **124** of FIGS. **1**

and/or **2**. In some examples, the display SoC **904** of FIG. **9** corresponds to the display SoC **126** of FIG. **1** and/or the SoC **490** of FIG. **4**. In some examples, the panel backlight **906** corresponds to the panel backlight **144** of FIGS. **1** and/or **2**.

[0130] During a first example operation **908** of the first data flow diagram **900**, the display SoC **904** power initializes the HPS **902**. For example, the sensor configurator **410** (FIG. **4**) directs the display SoC **904** to deliver power to the HPS **902**. During the first operation **908**, the display SoC **904** programs the HPS **902**. For example, the sensor configurator **410** programs the HPS **902** to configure one or more detection parameters of the HPS **902**, such as a presence confidence, an attention level, an attention confidence, a threshold of detection, etc., and/or a combination thereof. During the first operation **908**, the display SoC **904** queries the HPS **902** for an initial state of the HPS **902**. For example, the sensor interface **420** (FIG. **4**) queries the HPS **902** for a distance measurement of a present or instant object, such as the human **130** of FIG. **1**, whether the human **130** is present or not present, etc., and/or a combination thereof. At or prior to the end of the first operation **908**, the display panel **104**, **204** of FIGS. **1** and/or **2** is powered and fully operational by presenting video data, video frames, etc., from the main SoC **108** of FIG. **1**, to the human **130**. In some examples, irrespective of the human **130** present or absent, the display SoC **904** makes a state as if the human is present. In such examples, this ensures that the main SoC **108** and the display SoC **904** are in the same state of human presence information.

[0131] During a second example operation **910** of the first data flow diagram **900**, the display SoC **904** invokes the HPS **902** to start human absence detection. For example, the sensor interface **420** transmits a trigger command, a start command, etc., to the HPS **902** to begin detecting for absence of a user, such as the human **130**. During the second operation **910**, in response to starting human absence detection, the HPS **902** enters a human absence detection cycle.

[0132] During a third example operation **912** of the first data flow diagram **900**, the HPS **902** detects an absence of a user, such as the human **130**, and notifies human absence by asserting or generating a digital '1', and/or otherwise generating the HPS signal **132** of FIG. **1**. For example, the alert generator **460** (FIG. **4**) generates the HPS signal **132** to be indicative and/or otherwise representative of absence of the human **130**. During the third operation **912**, the HPS **902** transmits the HPS signal **132** to the display SoC **904** as the IRQ signal **134** of FIG. **1**. For example, the sensor interface **420** obtains the HPS signal **132** and the human detector **430** (FIG. **4**) determines that the human **130** is absent based on the HPS signal **132** being indicative of human absence.

[0133] During a fourth example operation **914** of the first data flow diagram **900**, the display SoC **904** queries the HPS **902** for detection parameters, such as presence confidence, attention level, attention confidence, a threshold of detection, etc., and/or a combination thereof. During the fourth operation **914**, in response to the query, the HPS **902** transmits the detection parameters to the display SoC **904**. For example, the sensor interface **420** queries the HPS **902** for the detection parameters and receive the detection parameters from the HPS **902**.

[0134] During a fifth example operation **916** of the first data flow diagram **900**, the display SoC **904** reduces a backlight power, intensity, etc., of the panel backlight **906** from a first level (e.g., 100% of peak power, 95% of peak power, etc.) to a second level (e.g., 20% of peak power, 15% of peak power, etc.). During the fifth operation **916**, the display SoC **904** starts and/or otherwise triggers a change state timer. For example, the human detector **430** (FIG. **4**) starts the change state timer to determine whether a time period for that the human **130** is absent satisfies a threshold (e.g., an absence threshold, a change state time threshold, etc.).

[0135] During the fifth operation **916**, in response to the change state timer elapsing, the display SoC **904** toggles the HPD signal, such as the HPD signal **140** of FIG. **1**, on a display interface, such as an HDMI interface. For example, in response to the HPS signal **132** being asserted and the HPD signal being asserted, the alert generator **460** de-asserts the HDL signal **142**, which, in some examples, is transmitted to the main SoC **108** of FIG. **1** via the display interface **110** of FIG. **1**. In such examples, in response to receiving the HDL signal **142**, the main SoC **108** ceases transmission

of video data to the display SoC **904** for presentation on the first display panel **104** of FIG. **1**. During the fifth operation **916**, in response to the change state timer elapsing, the display SoC **904** instructs the panel backlight **906** to turn or switch off. In response to turning off the panel backlight **906**, the first display panel **104** of FIG. **1** is turned off (e.g., fully or completely turned off) and the main SoC **108** has executed a power management policy and, thus, the main SoC **108** enters and/or otherwise transitions into an appropriate power state based on the power profile of the main SoC **108**.

[0136] During a sixth example operation **918** of the first data flow diagram **900**, the display SoC **904** transmits a reset sensor signal to the HPS **902** to detect human presence. For example, the sensor configurator **410** transmits a signal to the HPS **902** to reset the HPS **902** for detection of human presence instead of human absence. During the sixth operation **918**, the HPS **902** enters a human presence detection cycle.

[0137] During a seventh example operation **920** of the first data flow diagram **900**, the HPS **902** detects a presence of the human **130** and notifies the display SoC **904** of the human presence by toggling the HPS signal. For example, the alert generator **460** generates the HPS signal **132**. In such examples, the sensor interface **420** obtains the HPS signal **132** from the HPS **902** as the IRQ signal **134** in response to the HPS **902** detecting the human **130**. During the seventh operation **920**, the display SoC **904** toggles the HPD signal on the display interface in response to the human presence alert. For example, in response to a de-assertion of the HPS signal **132** and an assertion of the HPD signal **140**, the HDL **128** asserts the HDL signal **142** to be indicative of human presence.

[0138] During an eighth example operation **922** of the first data flow diagram **900**, the display SoC **904** switches on the backlight in response to the toggling of the HPD signal. During a ninth example operation **924** of the first data flow diagram **900**, the display SoC **904** sets the HPS **902** to detect for human absence.

[0139] FIGS. **10A-10B** depict a second example data flow diagram **1000** associated with the second example human detection system **202** of FIG. **2**, and/or, more generally, the second example computing environment **200** of FIG. **2**. The second data flow diagram **1000** is representative of example hardware logic, machine readable instructions, hardware implemented state machines, and/or any combination thereof for implementing an example HPS **1002**, an example display SoC **1004**, an example panel backlight **1006**, and an example main SoC **1008**. In some examples, the HPS **1002** of FIGS. **10A-10B** corresponds to the HPS **124** of FIG. **2**. In some examples, the display SoC **1004** of FIGS. **10A-10B** corresponds to the display SoC **126** of FIGS. **1** and/or **2**. In some examples, the panel backlight **1006** of FIGS. **10A-10B** corresponds to the panel backlight **144** of FIG. **2**. In some examples, the main SoC **1008** of FIGS. **10A-10B** corresponds to the main SoC **108** of FIGS. **1** and/or **2**.

[0140] During a first example operation **1010** of the second data flow diagram **1000**, the display SoC **1004** power initializes the HPS **1002**. For example, the sensor configurator **410** (FIG. **4**) directs the display SoC **1004** to deliver power to the HPS **1002**. During the first operation **1010**, the display SoC **1004** programs the HPS **1002**. For example, the sensor configurator **410** programs the HPS **1002** to configure one or more detection parameters of the HPS **1002**, such as a presence confidence, an attention level, an attention confidence, a threshold of detection, etc., and/or a combination thereof.

[0141] During the first operation **1010**, the display SoC **1004** queries the HPS **1002** for an initial state of the HPS **1002**. For example, the sensor interface **420** (FIG. **4**) queries the HPS **1002** for a distance measurement of a present or instant object, such as the human **130** of FIGS. **1** and/or **2**, whether the human **130** is present or not present, etc., and/or a combination thereof. At or prior to the end of the first operation **1010**, the second display panel **204** is powered and fully operational by presenting video data, video frames, etc., from the main SoC **1008** to the human **130**. In some examples, irrespective of the human **130** being present or absent, the display SoC **904** makes a state as if the human **130** is present. In such examples, this is to ensure that the main SoC **108** and the

display SoC **1108** are in the same state of human presence information.

[0142] During a second example operation **1012** of the second data flow diagram **1000**, the display SoC **1004** invokes the HPS **1002** to start human absence detection. For example, the sensor interface **420** transmits a trigger command, a start command, etc., to the HPS **1002** to begin detecting for absence of a user, such as the human **130**. During the second operation **1012**, in response to starting human absence detection, the HPS **1002** enters a human absence detection cycle.

[0143] During a third example operation **1014** of the second data flow diagram **1000**, the HPS **1002** detects an absence of a user, such as the human **130**, and notifies human absence by asserting or generating a digital '1', and/or otherwise generating the HPS signal **132** of FIGS. **1** and/or **2**. For example, the alert generator **460** (FIG. **4**) generates the HPS signal **132** to be indicative and/or otherwise representative of absence of the human **130**. During the third operation **1014**, the HPS **1002** transmits the HPS signal **132**, the IRQ signal **134**, etc., to the display SoC **1004**. For example, the sensor interface **420** obtains the HPS signal **132** and the human detector **430** (FIG. **4**) determines that the human **130** is absent based on the HPS signal **132** being indicative of human absence.

[0144] During a fourth example operation **1016** of the second data flow diagram **1000**, the main SoC **1008** queries the display SoC **1004** for detection parameters, such as presence confidence, attention level, attention confidence, a threshold of detection, etc., and/or a combination thereof. During the fourth operation **1016**, in response to the query, the display SoC **1004** transmits the detection parameters to the main SoC **1008**. For example, the sensor interface **420** queries the HPS **1002** for the detection parameters and receives the detection parameters from the HPS **1002**. In such examples, the display SoC **1004** transmits the received detection parameters to the main SoC **1008**.

[0145] During a fifth example operation **1018** of the second data flow diagram **1000**, the display SoC **1004** reduces a backlight power, intensity, etc., of the panel backlight **1006** from a first level (e.g., 100% of peak power, 95% of peak power, etc.) to a second level (e.g., 20% of peak power, 15% of peak power, etc.). For example, the command generator **450** (FIG. **4**) reduces a backlight power of the panel backlight **1006**. During the fifth operation **1018**, the display SoC **1004** notifies the main SoC **1008** over a display interface, such as the display interface **110** of FIGS. **1** and/or **2**, that a change in human presence has been detected. For example, in response to a de-assertion of the HPS signal **132**, the display SoC **1004** transmits an assertion of the HPD signal **140** of FIGS. **1** and/or **2** to the main SoC **1008** via the display interface **110**.

[0146] During a sixth example operation **1020** of the second data flow diagram **1000**, the main SoC **1008** communicates to a display subsystem, such as the VDC **120** of FIGS. **1** and/or **2**, to inform the change in the human presence. For example, the human detector **430** (FIG. **4**) informs the VDC **120** of the change in the human presence. During the sixth operation **1020**, the main SoC **1008** queries the display SoC **1004** for detection parameters of the HPS **1002** over an AUX channel of the display interface **110** of FIG. **2**. During the sixth operation **1020**, in response to the query, the display SoC **1004** returns the detection parameters to the display SoC **1004** over the AUX channel of the display interface **110**.

[0147] During the sixth operation **1020**, the main SoC **1008** performs host-side power policy framework action(s) to conserve system power. For example, the power controller **118** of FIG. **2** executes one or more power policy actions to conserve power of the main SoC **1008**, the display SoC **1004**, and/or, more generally, the second computing system **206** of FIG. **2**. In such examples, one or more power policy actions includes turning off a turbo performance option, reducing wireless fidelity (Wi-Fi) power (e.g., power delivered to Wi-Fi antenna (c) or radio(s)), and/or any other parameter, setting, etc., that affects power consumption and/or system performance. In some such examples, the main SoC **1008** executes the system wide power policy framework and enable systems that can put into a lower power mode, change a priority of tasks (e.g., computing tasks)

that are running, stop or trigger the pause of and/or presentation of audio and/or video data, transition the second computing system **206** into a power state (e.g., an Intel® S0ix state), etc. [0148] During the sixth operation **1020**, the main SoC **1008** sends a command over the display interface to shut down the display panel. For example, the VDC **120** of FIG. **2** generates a command and transmits the command to the display SoC **1004** via the display interface **110** to shut down and/or otherwise turn off the second display panel **204**.

[0149] During a seventh example operation **1022** of the second data flow diagram **1000**, the main SoC **1008** stops sending display frames to the display panel and transitions display controller into a power-saving mode. For example, the VDC **120** stops to cease transmitting display frames, video frames, etc., to the first display panel **104**. In such examples, the power controller **118** instructs the VDC **120** & main SOC **108** and its other component to move to a power-saving mode based on the host side power policy framework.

[0150] During an eighth example operation **1024** of the second data flow diagram **1000**, the display SoC **1004** switches off the backlight of the panel backlight **1006**. During the eighth operation **1024**, the display SoC **1004** performs display panel power optimization. For example, the display SoC **126** of FIG. **2** shuts down one or more computing processes, threads, etc., to transition the second display panel **204** into a power-saving mode.

[0151] During a ninth example operation **1026** of the second data flow diagram **1000**, the display SoC **1004** toggles the HPD signal. For example, the alert generator **460** generates the HPD signal **140** to be indicative of the second display panel **204** not being powered and/or otherwise operating in the power-saving mode.

[0152] During a tenth example operation **1028** of the second data flow diagram **1000**, the display SoC **1004** transmits a reset sensor signal to the HPS **1002** to detect human presence. For example, the sensor configurator **410** transmits a signal to the HPS **1002** to reset the HPS **1002** for detection of human presence instead of human absence. During the tenth operation **1028**, the HPS **1002** enters a human presence detection cycle.

[0153] During an eleventh example operation **1030** of the second data flow diagram **1000**, the HPS **1002** detects a presence of the human **130** and notifies the display SoC **1004** of the human presence by de-asserting the HPS signal. For example, the alert generator **460** generates the HPS signal **132** to be indicative of the human **130** being present proximate to the second display panel **204** and/or otherwise within range of the HPS **1002**. In such examples, the sensor interface **420** obtains the HPS signal **132**, the IRQ signal **134**, etc., from the HPS **1002** in response to the HPS **1002** detecting the human **130**. During the eleventh operation **1030**, the display SoC **1004** toggles the HPD signal **140** of FIG. **2** on the display interface in response to the human presence alert.

[0154] During a twelfth example operation **1032** of the second data flow diagram **1000**, the main SoC **1008** communicates to the display subsystem, such as the VDC **120** of FIGS. **1** and/or **2**, to inform the change in the human presence. For example, the human detector **430** informs the VDC **120** of the change in the human presence over the AUX channel of the display interface **110**. During the twelfth operation **1032**, the main SoC **1008** queries the display SoC **1004** for detection parameters of the HPS **1002**. During the twelfth operation **1032**, in response to the query, the display SoC **1004** returns the detection parameters to the display SoC **1004** via the display interface, such as the display interface **110** of FIGS. **1** and/or **2**.

[0155] During the twelfth operation **1032**, the main SoC **1008** performs host-side power policy framework action(s) to transition from a power-saving mode to an active power mode. For example, the power controller **118** executes one or more power policy actions to enable an active power mode of the main SoC **1008**, the display SoC **1004**, and/or, more generally, the second computing system **206** of FIG. **2**. In such examples, the one or more power policy actions includes turning on a turbo performance option, increasing wireless fidelity (Wi-Fi) power (e.g., power delivered to Wi-Fi antenna (c) or radio(s)), and/or any other parameter, setting, etc., that affects power consumption and/or system performance.

[0156] During a thirteen example operation **1034** of the second data flow diagram **1000**, the display SoC **1004** sends a command and panel configuration information over the display interface to restart the display panel. For example, the VDC **120** of FIG. **2** generates a command and transmits the command to the display SoC **1004** via the display interface **110** to restart and/or otherwise turn on the second display panel **204**. In such examples, the display SoC **1004** performs routine steps to return the display panel configuration to the main SoC **1008**. During a fourteenth example operation **1036**, the display SoC **1004** switches on the backlight of the panel backlight **1006**.

[0157] During a fifteenth example operation **1038** of the second data flow diagram **1000**, the main SoC **1008** starts sending display frames to the display SoC **1004**. For example, the VDC **120** transmits display frames to the display SoC **1004** for presentation on the second display panel **204**.

[0158] During a sixteenth example operation **1040** of the second data flow diagram **1000**, the display SoC **1004** performs a display panel power-on and setups the display panel. For example, the display SoC **1004** to perform a display panel power-on operation and setup the second display panel **204** for presentation of the video frames from the main SoC **1008**. During a seventeenth example operation **1042** of the second data flow diagram **1000**, the display SoC **1004** sets the HPS **1002** to detect for human absence.

[0159] FIGS. **11A-11B** depict a third example data flow diagram **1100** associated with the SoC **304** of FIG. **3**, and/or, more generally, the third computing system **300** of FIG. **3**. The third data flow diagram **1100** is representative of example hardware logic, machine readable instructions, hardware implemented state machines, and/or any combination thereof for implementing an HPS **1102**, an ISH **1104**, a main SoC **1106**, a display SoC **1108**, a panel backlight **1110**. In some examples, the HPS **1102** of FIGS. **11A-11B** corresponds to the HPS **306** of FIG. **3**. In some examples, the ISH **1104** of FIGS. **11A-11B** corresponds to the ISH **320** of FIG. **3**. In some examples, the main SoC **1106** of FIGS. **11A-11B** corresponds to the SoC **304** of FIG. **3**. In some examples, the display SoC **1108** of FIGS. **11A-11B** corresponds to an SoC included in the third display panel **302** of FIG. **3**. In some examples, the panel backlight **1110** of FIGS. **11A-11B** correspond to a panel backlight included in the third display panel **302** of FIG. **3**, such as the panel backlight **144** of FIGS. **1** and/or **2**.

[0160] During a first example operation **1112** of the third data flow diagram **1100**, the ISH **1104** power initializes the HPS **1102**, programs the HPS **1102**, and queries the HPS **1102** for an initial state of the HPS **1102**. At or prior to the end of the first operation **1112**, the third display panel **302** is powered and fully operational by presenting video data, video frames, etc., from the main SoC **1106** to the human **130**.

[0161] During a second example operation **1114** of the third data flow diagram **1100**, the ISH **1104** invokes the HPS **1102** to start human absence detection to cause the HPS **1102** to enter a human absence detection cycle. During a third example operation **1116** of the third data flow diagram **1100**, the HPS **1102** detects an absence of a user, such as the human **130**, and notifies human absence by toggling and/or otherwise generating the HPS signal **132** of FIGS. **1** and/or **2** via the HPS interface **310** of FIG. **3**.

[0162] During a fourth example operation **1118** of the third data flow diagram **1100**, the ISH **1104** queries the HPS **1102** for detection parameters, and, in response to the query, the HPS **1102** transmits the detection parameters to the ISH **1104**. During a fifth example operation **1120** of the third data flow diagram **1100**, the ISH **1104** notifies the main SoC **1106** over a bus, such as the bus **322** of FIG. **3**, that a change in human presence has been detected.

[0163] During a sixth example operation **1122** of the third data flow diagram **1100**, the main SoC **1106** performs host-side power policy framework action(s) to conserve system power. During a seventh operation **1124** of the third data flow diagram **1100**, the main SoC **1106** sends a command over a display interface, such as the display interface **308** of FIG. **3**, to shut down the third display panel **302**. During the seventh operation **1124**, in response to the command, the display SoC **1108** switches off the panel backlight **1110**.

[0164] During an eighth example operation **1126** of the third data flow diagram **1100**, the main SoC **1106** stops sending display frames to the third display panel **302** and transitions a display controller, such as the VDC **120** of FIG. **3**, into a power-saving mode. During the eighth operation **1126**, the display SoC **1108** performs a display panel power optimization.

[0165] During a ninth example operation **1128** of the third data flow diagram **1100**, the ISH **1104** resets the sensor signal of the HPS **1102** to detect for human presence. During the ninth operation **1128**, in response to the reset, the HPS **1102** enters a human presence detection cycle.

[0166] During a tenth example operation **1130** of the third data flow diagram **1100**, the HPS **1102** detects the human **130**, and transmits a notification of a detection of human presence to the ISH **1104** by toggling the HPS signal of the HPS **1102**. During the tenth operation **1130**, the ISH **1104** notifies the main SoC **1106** of the change in human presence over a bus, such as the bus **322** of FIG. **3**.

[0167] During an eleventh example operation **1132** of the third data flow diagram **1100**, the main SoC **1106** performs host-side power policy framework action(s) to transition the main SoC **1106** to an active power mode. During a twelfth example operation **1134** of the third data flow diagram **1100**, the main SoC **1106** sends a command and panel configuration information to the display SoC **1108** over a display interface, such as the display interface **308** of FIG. **3**. During the twelfth operation **1134**, the main SoC **1106** starts sending display frames to the third display panel **302**.

[0168] During a thirteenth example operation **1136** of the third data flow diagram **1100**, the display SoC **1108** switches on the panel backlight **1110** and performs a display power turn-on and setup of the third display panel **302**. During a fourteenth example operation **1138** of the third data flow diagram **1100**, the ISH **1104** returns to set the HPS **1102** for human absence detection.

[0169] Advantageously, in some examples, the first human detection system **102** of FIG. **1**, the second human detection system **202** of FIG. **2**, and/or the third human detection system **400** of FIG. **4** adjusts a power state of a display device, such as the first display panel **104** of FIG. **1** and/or the second display panel **204** of FIG. **2**, without interfacing with an ISH, such as the ISH **320** of FIG. **3**. Advantageously, the first human detection system **102** of FIG. **1**, the second human detection system **202** of FIG. **2**, and/or the third human detection system **400** of FIG. **4** are improvement(s) of the third computing system **300** of FIG. **3** by effectuating display power management with example hardware, software, and/or firmware included in and/or executing on the display panel and, in some examples, without interfacing with a host OS, a main SoC interface, etc.

[0170] FIG. **12** is a flowchart representative of example machine readable instructions **1200** that may be executed to implement the first human detection system **102** of FIG. **1**, the human detection controller **405** of FIG. **4**, and/or, more generally, the third human detection system **400** of FIG. **4**, to invoke hardware in the first example display panel **104** of FIG. **1** to adjust a power state of the first display panel **104**. The machine readable instructions **1200** of FIG. **12** begin at block **1202**, at which the third human detection system **400** initializes a human presence sensor in a display device. For example, the sensor configurator **410** (FIG. **4**) power initializes the HPS **124** of FIG. **1**. In such examples, the sensor configurator **410** programs the HPS **124** using one or more detection parameters.

[0171] At block **1204**, the third human detection system **400** obtains detection data from the human presence sensor. For example, the sensor interface **420** (FIG. **4**) obtains at least one of the HPS signal **132** or the HPS data **138** from the HPS **124**. In such examples, the sensor interface **420** obtains an initial or current state of the HPS **124**, such as whether the human **130** is detected, a distance of the human **130** from the first display panel **104**, etc.

[0172] At block **1206**, the third human detection system **400** determines whether a change in human presence proximate the display device is detected. For example, the HPS **124** determines whether a change in the presence of the human **130** is detected. In such examples, the HPS **124** determines that the change in the presence of the human **130** is detected when the human **130** steps away from the first display panel **104** and is no longer proximate the first display panel **104**, returns

to the first display panel **104** within a range (e.g., a detection range) of the HPS **124** and is proximate the first display panel **104**, etc.

[0173] If, at block **1206**, the third human detection system **400** determines that a change in human presence proximate the display device is not detected, control returns to block **1204** to obtain detection data from the human presence sensor. If, at block **1206**, the third human detection system **400** determines that a change in human presence proximate the display device is detected, then, at block **1208**, the third human detection system **400** generates an alert indicative of a change in human presence. For example, the HPS **124** generates the HPS signal **132** based on whether the human **130** is present. In such examples, the HPS **124** de-asserts the HPS signal **132** in response to detecting the human **130** and asserts the HPS signal **132** in response to not detecting the human **130**. The human detector **430** (FIG. 4) determines that the HPS **124** detects the human **130** based on the HPS signal **132**. In such examples, the human detector **430** determines that a de-assertion of the HPS signal **132** is indicative of the HPS **124** detecting the human **130** and an assertion of the HPS signal **132** is indicative of the HPS **124** not detecting the human **130**.

[0174] At block **1210**, the third human detection system **400** facilitates timer operation. For example, in response to receiving the HPS signal **132** from the HPS **124**, the human detector **430** triggers a power state change timer to begin in response to the alert.

[0175] At block **1212**, the third human detection system **400** determines whether the timer has elapsed without an additional change in human presence. For example, the human detector **430** determines that the power state change timer has elapsed without an additional change in the human presence, such as the HPS **124** first determining an absence of the human **130** and then determine a presence of the human **130** before the power state change timer has elapsed.

[0176] If, at block **1212**, the third human detection system **400** determines that the timer has not elapsed and/or an additional change in human presence has been detected, control returns to block **1206** to determine whether a change in human presence proximate the display device is detected. If, at block **1212**, the third human detection system **400** determines that the timer has elapsed without an additional change in human presence being detected, then, at block **1214**, the third human detection system **400** determines whether communication of the display device with host hardware has been detected. For example, the display SoC **126** determines that the first display panel **104** is in communication with the main SoC **108** of FIG. 1 via the display interface **110** of FIG. 1. In such examples, the alert generator **460** generates the HPD signal **140** of FIG. 1 based on the detection.

[0177] If, at block **1214**, the third human detection system **400** determines that the display device is not in communication with the host hardware, control proceeds to block **1218** to determine whether to continue detecting for human presence proximate the display device. If, at block **1214**, the third human detection system **400** determines that the display device is in communication with the host hardware, then, at block **1216**, the third human detection system **400** invokes hardware in the display device to adjust a power state of the display device. For example, the display SoC **126** changes a brightness of the panel backlight **144**, turns on or off the first display panel **104**, generate the HDL signal **142** to be indicative of the change in human presence, etc. In such examples, the alert generator **460** asserts the HDL signal **142** in response to a de-assertion of the HPS signal **132** and an assertion of the HPD signal **140**. In some examples, the HPS **124** generates the IRQ signal **134** to be transmitted to the display SoC **126**. In some such examples, the alert generator **460** de-asserts the HDL signal **142** in response to an assertion of the HPS signal **132** and an assertion of the HPD signal **140**. In some examples, in response to receiving the HDL signal **142**, the VDC **120** of FIG. 1 determines to cease or begin transmitting video frames to the first display panel **104**.

[0178] At block **1218**, the third human detection system **400** determines whether to continue detecting for human presence proximate the display device. For example, the sensor interface **420** determines whether to continue monitoring the HPS **124** for a change in human presence. If, at block **1218**, the third human detection system **400** determines to continue detecting for human

presence proximate the display device, control returns to block **1206** to determine whether a change in human presence proximate the display device has been detected. If, at block **1218**, the third human detection system **400** determines not to continue detecting for human presence proximate the display device, the machine readable instructions **1200** of FIG. **12** conclude.

[0179] FIG. **13** is a flowchart representative of example machine readable instructions **1300** that may be executed to implement the human detection controller of **405** of FIG. **4**, and/or, more generally, the third human detection system **400** of FIG. **4**, to invoke hardware in the first display panel **104** of FIG. **1** and/or the main SoC **108** of FIG. **1** to adjust a power state of the first display panel **104**. The machine readable instructions **1300** of FIG. **13** begin at block **1302**, at which the third human detection system **400** initializes a human presence sensor in a display device. For example, the sensor configurator **410** (FIG. **4**) power initializes the HPS **124** of FIGS. **1** and/or **2**. In such examples, the sensor configurator **410** programs the HPS **124** using one or more detection parameters.

[0180] At block **1304**, the third human detection system **400** determines whether a human is detected proximate the display device. For example, the HPS **124** determines whether the human **130** is within a detection range of the HPS **124**, and/or, more generally, the first display panel **104**.

[0181] If, at block **1304**, the third human detection system **400** determines that the human is not detected proximate the display device, then, at block **1306**, the third human detection system **400** transmits a signal indicative of human absence to hardware in the display device to turn off a backlight and trigger a timer. For example, the HPS **124** transmits the HPS signal **132** to the display SoC **126** as the IRQ signal **134**. In such examples, the sensor interface **420** (FIG. **4**) receives the IRQ signal **134** and the human detector **430** (FIG. **4**) determines whether the human **130** is absent based on the IRQ signal **134**. The human detector **430** triggers a power state change timer based on the IRQ signal **134**. In some such examples, in response to the human detector **430** determining that the human **130** is not present, the command generator **450** (FIG. **4**) directs the first display panel **104** to turn off the backlight of the first display panel **104**.

[0182] At block **1308**, the third human detection system **400** determines whether the timer has elapsed without an additional change in human presence. For example, the human detector **430** determines that the power state change timer has elapsed without an additional change in the human presence, such as the HPS **124** first determining an absence of the human **130** and then determining a presence of the human **130** before the power state change timer has elapsed.

[0183] If, at block **1308**, the third human detection system **400** determines that the timer has not elapsed and/or an additional change in human presence has been detected, control returns to block **1304** to determine whether a change in human presence proximate the display device is detected. If, at block **1308**, the third human detection system **400** determines that the timer has elapsed without an additional change in human presence being detected, then, at block **1310**, the third human detection system **400** determines whether the display device is connected to host hardware. For example, the human detector **430** determines that the first display panel **104** is in communication with the main SoC **108** of FIG. **1** via the display interface **110** of FIG. **1**. In such examples, the alert generator **460** generates the HPD signal **140** of FIG. **1** based on the detection.

[0184] If, at block **1310**, the third human detection system **400** determines that the display device is not connected to host hardware, control proceeds to block **1320** to determine whether to continue detecting for human presence proximate the display device. If, at block **1310**, the third human detection system **400** determines that the display device is connected to host hardware, then, at block **1312**, the third human detection system **400** transmits a signal indicative of human absence to the host hardware to cease transmission of display data. For example, the alert generator **460** transmits the HDL signal **142** to the main SoC **108** to invoke the VDC **120** to stop transmitting video frames to the first display panel **104**. In response to transmitting the signal indicative of human absence to the host hardware to cease transmission of display data at block **1312**, control proceeds to block **1320** to determine whether to continue detecting for human presence proximate

the display device.

[0185] If, at block **1304**, the third human detection system **400** determines that the human is detected proximate the display device, control proceeds to block **1314** to transmit a signal indicative of human detection to hardware in the display device to turn on a backlight. For example, the command generator **450** instructs the display SoC **126** to turn on the backlight of the first display panel **104**.

[0186] At block **1316**, the third human detection system **400** determines whether the display device is connected to host hardware. For example, the human detector **430** determines that the first display panel **104** is in communication with the main SoC **108** of FIG. **1** via the display interface **110** of FIG. **1**. In such examples, the alert generator **460** generates the HPD signal **140** of FIG. **1** based on the detection.

[0187] If, at block **1316**, the third human detection system **400** determines that the display device is not connected to host hardware, control proceeds to block **1320** to determine whether to continue detecting for human presence proximate the display device. If, at block **1316**, the third human detection system **400** determines that the display device is connected to host hardware, then, at block **1318**, the third human detection system **400** transmits a signal indicative of human presence to the host hardware to trigger transmission of display data. For example, the alert generator **460** transmits the HDL signal **142** to the main SoC **108** to invoke the VDC **120** to begin transmitting video frames to the first display panel **104**.

[0188] At block **1320**, the third human detection system **400** determines whether to continue detecting for human presence proximate the display device. For example, the sensor interface **420** determines whether to continue monitoring the HPS **124** for a change in human presence. If, at block **1320**, the third human detection system **400** determines to continue detecting for human presence proximate the display device, control returns to block **1304** to determine whether a human is detected proximate the display device. If, at block **1320**, the third human detection system **400** determines not to continue detecting for human presence proximate the display device, the machine readable instructions **1300** of FIG. **13** conclude.

[0189] FIG. **14** is a flowchart representative of example machine readable instructions **1400** that may be executed to implement the human detection controller **405** of FIG. **4**, and/or, more generally, the third human detection system **400** of FIG. **4**, to invoke the main SoC **108** of FIGS. **1** and/or **2** to adjust a power state of the second display panel **204** of FIG. **2**. The machine readable instructions **1400** of FIG. **14** begin at block **1402**, at which the third human detection system **400** initializes a human presence sensor in a display device. For example, the sensor configurator **410** (FIG. **4**) power initializes the HPS **124** of FIGS. **1** and/or **2**. In such examples, the sensor configurator **410** configures the HPS **124** using one or more detection parameters.

[0190] At block **1404**, the third human detection system **400** determines whether a change in human presence is detected. For example, the HPS **124** generates the HPS signal **132** based on determining whether the human **130** of FIGS. **1** and/or **2** is within a detection range of the HPS **124**, and/or, more generally, the second display panel **204**. In such examples, the human detector **430** (FIG. **4**) determines that the human **130** is within the detection range based on the HPS signal **132**.

[0191] If, at block **1404**, the third human detection system **400** determines that a change in human presence is not detected, control proceeds to block **1418** to determine whether to continue detecting for human presence proximate the display device. If, at block **1404**, the third human detection system **400** determines that a change in human presence is detected, then, at block **1406**, the third human detection system **400** generates an alert indicative of a change in human presence proximate the display device. For example, the alert generator **460** (FIG. **4**) generates the HPD signal **140** based on the detection of the human **130**. In such examples, the alert generator **460** asserts the HPD signal **140** in response to detecting the human **130** and de-asserts the HPD signal **140** in response to not detecting the human **130**.

[0192] At block **1408**, the third human detection system **400** queries the display device for detection parameters associated with the human presence sensor. For example, the main SoC **108** queries the display SoC **126** for one or more detection parameters associated with the HPS **124**. In such examples, the sensor configurator **410** returns the one or more detection parameters to the main SoC **108**.

[0193] At block **1410**, the third human detection system **400** transmits detection data including the alert and the detection parameters to host hardware. For example, the alert generator **460** transmits detection data including at least one of the display data **127** or the HPD signal **140** to the main SoC **108** via the display interface **110**.

[0194] At block **1412**, the third human detection system **400** compares the detection data to power profile(s). For example, the power profiler **440** (FIG. 4) compares the HPD signal **140**, the one or more detection parameters, etc., and/or a combination thereof, to the power profile(s) **474** (FIG. 4). In other examples, the main SoC **108** compares the HPD signal **140**, the one or more detection parameters, etc., and/or a combination thereof, to the power profile(s) **474** (FIG. 4).

[0195] At block **1414**, the third human detection system **400** determines whether to change a display device power state based on the comparison. For example, the power profiler **440** determines to reduce a brightness, turn off the backlight, etc., of the second display panel **204** based on the comparison. In other examples, the power profiler **440** determines to increase a brightness, turn on the backlight, etc., of the second display panel **204** based on the comparison. In yet other examples, the main SoC **108** determines to change the display device power state based on the comparison.

[0196] If, at block **1414**, the third human detection system **400** determines not to change the display device power state based on the comparison, control returns to block **1404** to determine whether a change in human presence has been detected. If, at block **1414**, the third human detection system **400** determines to change the display device power state based on the comparison, then, at block **1416**, the third human detection system **400** invokes the display device to change the display device power state. For example, the command generator **450** instructs the second display panel **204** to change a display device power state. In other examples, the main SoC **108** instructs the second display panel **204** to change the display device power state.

[0197] At block **1418**, the third human detection system **400** determines whether to continue detecting for human presence proximate the display device. For example, the sensor interface **420** determines whether to continue monitoring the HPS **124** for a change in human presence. If, at block **1418**, the third human detection system **400** determines to continue detecting for human presence proximate the display device, control returns to block **1404** to determine whether a change in human presence proximate the display device has been detected. If, at block **1418**, the third human detection system **400** determines not to continue detecting for human presence proximate the display device, the machine readable instructions **1400** of FIG. 14 conclude.

[0198] FIG. 15 is a block diagram of an example processor platform **1500** structured to execute the instructions of FIGS. 9-14 to implement the example human detection controller **405** of FIG. 4. The processor platform **1500** can be, for example, a display panel, a personal computer, a workstation, or any other type of computing device. In some examples, the processor platform **1500** implements the first display panel **104** of FIG. 1 and/or portion(s) thereof. In some examples, the processor platform **1500** implements the second display panel **204** of FIG. 2 and/or portion(s) thereof.

[0199] The processor platform **1500** of the illustrated example includes a processor **1512**. The processor **1512** of the illustrated example is hardware. For example, the processor **1512** can be implemented by one or more integrated circuits, logic circuits, microprocessors, GPUs, DSPs, or controllers from any desired family or manufacturer. The hardware processor may be a semiconductor based (e.g., silicon based) device. In this example, the processor **1512** implements the example sensor configurator **410**, the example human detector **430**, the example power profiler **440**, the example command generator **450**, and the example alert generator **460** of FIG. 4.

[0200] The processor **1512** of the illustrated example includes a local memory **1513** (e.g., a cache). The processor **1512** of the illustrated example is in communication with a main memory including a volatile memory **1514** and a non-volatile memory **1516** via a bus **1518**. The volatile memory **1514** may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®) and/or any other type of random access memory device. The non-volatile memory **1516** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **1514**, **1516** is controlled by a memory controller.

[0201] The processor platform **1500** of the illustrated example also includes an interface circuit **1520**. The interface circuit **1520** may be implemented by any type of interface standard, such as an Ethernet interface, a universal serial bus (USB), a Bluetooth® interface, a near field communication (NFC) interface, a PCI express interface, an I2C interface, and/or a SPI or other serial interface. In this example, the interface circuit **1520** implements the example sensor interface **420** of FIG. 4.

[0202] In the illustrated example, one or more input devices **1522** are connected to the interface circuit **1520**. The input device(s) **1522** permit(s) a user to enter data and/or commands into the processor **1512**. The input device(s) **1522** can be implemented by, for example, an HPS, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, isopoint, and/or a voice recognition system.

[0203] One or more output devices **1524** are also connected to the interface circuit **1520** of the illustrated example. The output devices **1524** can be implemented, for example, by display devices (e.g., an LED, an OLED, an LCD, a CRT display, an IPS display, a touchscreen, etc.), a tactile output device, a printer and/or speaker. The interface circuit **1520** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip and/or a graphics driver processor.

[0204] The interface circuit **1520** of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) via a network **1526**. The communication can be via, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, etc.

[0205] The processor platform **1500** of the illustrated example also includes one or more mass storage devices **1528** for storing software and/or data. Examples of such mass storage devices **1528** include floppy disk drives, hard drive disks, compact disk drives, Blu-ray disk drives, redundant array of independent disks (RAID) systems, and digital versatile disk (DVD) drives. In this example, the one or more mass storage devices **1528** implement the example database **470**, the example sensor configuration(s) **472**, and the example power profile(s) **474** of FIG. 4.

[0206] The machine executable instructions **1532** of FIGS. 9-14 may be stored in the mass storage device **1528**, in the volatile memory **1514**, in the non-volatile memory **1516**, and/or on a removable non-transitory computer readable storage medium such as a CD or DVD.

[0207] FIG. 16 is a block diagram of an example processor platform **1600** structured to execute the instructions of FIGS. 9-14 to implement the example human detection controller **405** of FIG. 4. The processor platform **1600** can be, for example, a display panel, a personal computer, a workstation, or any other type of computing device. In some examples, the processor platform **1600** implements the first display panel **104** of FIG. 1 and/or portion(s) thereof. In some examples, the processor platform **1600** implements the second display panel **204** of FIG. 2 and/or portion(s) thereof.

[0208] The processor platform **1600** of the illustrated example includes a processor **1612**. The processor **1612** of the illustrated example is hardware. For example, the processor **1612** can be implemented by one or more integrated circuits, logic circuits, microprocessors, GPUs, DSPs, or controllers from any desired family or manufacturer. The hardware processor may be a

semiconductor based (e.g., silicon based) device.

[0209] The processor **1612** of the illustrated example includes a local memory **1613** (e.g., a cache). The processor **1612** of the illustrated example is in communication with a main memory including a volatile memory **1614** and a non-volatile memory **1616** via a bus **1618**. The volatile memory **1614** may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®) and/or any other type of random access memory device. The non-volatile memory **1616** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **1614**, **1616** is controlled by a memory controller.

[0210] The processor platform **1600** of the illustrated example also includes an interface circuit **1620**. The interface circuit **1620** may be implemented by any type of interface standard, such as an Ethernet interface, a universal serial bus (USB), a Bluetooth® interface, a near field communication (NFC) interface, a PCI express interface, an I2C interface, and/or a SPI or other serial interface. In this example, the interface circuit **1620** implements the example sensor interface **420** of FIG. 4.

[0211] In the illustrated example, one or more input devices **1622** are connected to the interface circuit **1620**. The input device(s) **1622** permit(s) a user to enter data and/or commands into the processor **1612**. The input device(s) **1622** can be implemented by, for example, an HPS, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, isopoint, and/or a voice recognition system. In this example, the input device(s) **1622** implement the example sensor configurator **410**, the example human detector **430**, the example power profiler **440**, the example command generator **450**, and the example alert generator **460** of FIG. 4. Additionally or alternatively, the input device(s) **1622** may implement and/or otherwise include the sensor interface **420**, the database **270**, the sensor configuration(s) **472**, and/or the power profile(s) **474**. In some examples, the processor platform **1600** of the illustrated example is representative of a class, a genre, a spectrum, etc., of the input device(s) **1622** that also includes an interface circuit, such as the interface circuit **1620**.

[0212] One or more output devices **1624** are also connected to the interface circuit **1620** of the illustrated example. The output devices **1624** can be implemented, for example, by display devices (e.g., an LED, an OLED, an LCD, a CRT display, an IPS display, a touchscreen, etc.), a tactile output device, a printer and/or speaker. The interface circuit **1620** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip and/or a graphics driver processor.

[0213] The interface circuit **1620** of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) via a network **1626**. The communication can be via, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, etc.

[0214] The processor platform **1600** of the illustrated example also includes one or more mass storage devices **1628** for storing software and/or data. Examples of such mass storage devices **1628** include floppy disk drives, hard drive disks, compact disk drives, Blu-ray disk drives, redundant array of independent disks (RAID) systems, and digital versatile disk (DVD) drives. In this example, the one or more mass storage devices **1628** implement the example database **470**, the example sensor configuration(s) **472**, and the example power profile(s) **474** of FIG. 4.

[0215] The machine executable instructions **1632** of FIGS. 9-14 may be stored in the mass storage device **1628**, in the volatile memory **1614**, in the non-volatile memory **1616**, and/or on a removable non-transitory computer readable storage medium such as a CD or DVD.

[0216] FIG. 17 is a block diagram of an example processor platform **1700** structured to execute the instructions of FIGS. 9-14 to implement the example human detection controller **405** of FIG. 4. The

processor platform **1700** can be, for example, a display panel, a personal computer, a workstation, or any other type of computing device. In some examples, the processor platform **1700** implements the first display panel **104** of FIG. **1** and/or portion(s) thereof. In some examples, the processor platform **1700** implements the second display panel **204** of FIG. **2** and/or portion(s) thereof. [0217] The processor platform **1700** of the illustrated example includes a processor **1712**. The processor **1712** of the illustrated example is hardware. For example, the processor **1712** can be implemented by one or more integrated circuits, logic circuits, microprocessors, GPUs, DSPs, or controllers from any desired family or manufacturer. The hardware processor may be a semiconductor based (e.g., silicon based) device.

[0218] The processor **1712** of the illustrated example includes a local memory **1713** (e.g., a cache). The processor **1712** of the illustrated example is in communication with a main memory including a volatile memory **1714** and a non-volatile memory **1716** via a bus **1718**. The volatile memory **1714** may be implemented by Synchronous Dynamic Random Access Memory (SDRAM), Dynamic Random Access Memory (DRAM), RAMBUS® Dynamic Random Access Memory (RDRAM®) and/or any other type of random access memory device. The non-volatile memory **1716** may be implemented by flash memory and/or any other desired type of memory device. Access to the main memory **1714**, **1716** is controlled by a memory controller.

[0219] The processor platform **1700** of the illustrated example also includes an interface circuit **1720**. The interface circuit **1720** may be implemented by any type of interface standard, such as an Ethernet interface, a universal serial bus (USB), a Bluetooth® interface, a near field communication (NFC) interface, a PCI express interface, an I2C interface, and/or a SPI or other serial interface. In this example, the interface circuit **1720** implements the example sensor interface **420** of FIG. **4**. In some examples, the processor platform **1700** of the illustrated example is representative of a class, a genre, a spectrum, etc., of the input device(s) **1722** that also includes an interface circuit, such as the interface circuit **1720**.

[0220] In the illustrated example, one or more input devices **1722** are connected to the interface circuit **1720**. The input device(s) **1722** permit(s) a user to enter data and/or commands into the processor **1712**. The input device(s) **1722** can be implemented by, for example, an HPS, an audio sensor, a microphone, a camera (still or video), a keyboard, a button, a mouse, a touchscreen, a track-pad, a trackball, isopoint, and/or a voice recognition system.

[0221] One or more output devices **1724** are also connected to the interface circuit **1720** of the illustrated example. The output devices **1724** can be implemented, for example, by display devices (e.g., an LED, an OLED, an LCD, a CRT display, an IPS display, a touchscreen, etc.), a tactile output device, a printer and/or speaker. The interface circuit **1720** of the illustrated example, thus, typically includes a graphics driver card, a graphics driver chip and/or a graphics driver processor. In this example, the output device(s) **1724** implement the example sensor configurator **410**, the example human detector **430**, the example power profiler **440**, the example command generator **450**, and the example alert generator **460** of FIG. **4**. Additionally or alternatively, the output device(s) **1724** may implement and/or otherwise include the sensor interface **420**, the database **270**, the sensor configuration(s) **472**, and/or the power profile(s) **474**. In some examples, the processor platform **1700** of the illustrated example is representative of a class, a genre, a spectrum, etc., of the output device(s) **1724** that also includes an interface circuit, such as the interface circuit **1720**.

[0222] The interface circuit **1720** of the illustrated example also includes a communication device such as a transmitter, a receiver, a transceiver, a modem, a residential gateway, a wireless access point, and/or a network interface to facilitate exchange of data with external machines (e.g., computing devices of any kind) via a network **1726**. The communication can be via, for example, an Ethernet connection, a digital subscriber line (DSL) connection, a telephone line connection, a coaxial cable system, a satellite system, a line-of-site wireless system, a cellular telephone system, etc.

[0223] The processor platform **1700** of the illustrated example also includes one or more mass

storage devices **1728** for storing software and/or data. Examples of such mass storage devices **1728** include floppy disk drives, hard drive disks, compact disk drives, Blu-ray disk drives, redundant array of independent disks (RAID) systems, and digital versatile disk (DVD) drives. In this example, the one or more mass storage devices **1728** implement the example database **470**, the example sensor configuration(s) **472**, and the example power profile(s) **474** of FIG. 4.

[0224] The machine executable instructions **1732** of FIGS. 9-14 may be stored in the mass storage device **1728**, in the volatile memory **1714**, in the non-volatile memory **1716**, and/or on a removable non-transitory computer readable storage medium such as a CD or DVD.

[0225] From the foregoing, it will be appreciated that example systems, methods, apparatus, and articles of manufacture have been disclosed for display panel human presence detection. The disclosed systems, methods, apparatus, and articles of manufacture improve the efficiency of using a computing device to reduce power consumption and utilization of computational resources (e.g., processor(s), memory, storage, etc.) when a user steps away from the computing device. The disclosed systems, methods, apparatus, and articles of manufacture improve the efficiency of using a computing device by offloading human detection computing tasks from a host OS of host hardware to hardware included in a display panel and, thus, reduce the utilization of host hardware resources to execute the human detection computing tasks. The disclosed systems, methods, apparatus, and articles of manufacture are accordingly directed to one or more improvement(s) in the functioning of a computer.

[0226] Example methods, apparatus, systems, and articles of manufacture for display panel human detection are disclosed herein. Further examples and combinations thereof include the following:

[0227] Example 1 includes a display device comprising a sensor to generate a detection signal based on whether a human is detected, and logic circuitry to generate a control signal to cause an adjustment of a power state of the display device based on the detection signal. [0228] Example 2 includes the display device of example 1, wherein the logic circuitry is to generate the control signal to reduce a backlight intensity of the display device or turn off the display device. [0229] Example 3 includes the display device of example 1, wherein the logic circuitry is to determine that the detection signal is indicative of a presence of the human proximate the display device, and determine that a connection status signal indicates the display device is in communication with host hardware. [0230] Example 4 includes the display device of example 3, wherein the control signal is to adjust the power state of the display device from a disabled state to an enabled state. [0231] Example 5 includes the display device of example 1, wherein the logic circuitry is to determine that the detection signal is indicative of an absence of the human proximate the display device, and determine that a connection status signal indicates the display device is in communication with host hardware. [0232] Example 6 includes the display device of example 5, wherein the control signal is to adjust the power state of the display device from an enabled state to a disabled state. [0233] Example 7 includes the display device of example 1, further including display device hardware in circuit with the sensor and the logic circuitry, the display device hardware to assert a connection status signal in response to detecting that the display device is in communication with host hardware, and de-assert the connection status signal in response to not detecting that the display device is in communication with the host hardware. [0234] Example 8 includes the display device of example 1, wherein the control signal is a first control signal, further including display device hardware in circuit with the sensor and the logic circuitry, the display device hardware to generate a connection status signal in response to detecting that the display device is in communication with host hardware, in response to obtaining the control signal, trigger a timer, in response to the timer elapsing, turn off the display device, and transmit a second control signal to the host hardware to cease transmission of display data from the host hardware to the display device. [0235] Example 9 includes the display device of example 8, wherein the display device hardware is to turn off the display device in response to the timer elapsing without a change in the detection signal. [0236] Example 10 includes the display device of example 1, further including display device hardware in

circuit with the sensor, the display device hardware to configure the sensor based on detection parameters, and transmit at least one of the detection parameters or the detection signal to host hardware in communication with the display device hardware over a communication channel of a display interface. [0237] Example 11 includes the display interface of example 10, wherein the communication channel is an auxiliary channel of a display interface. [0238] Example 12 includes the display device of example 1, wherein the sensor is to generate the detection signal based on whether the human is detected proximate the display device. [0239] Example 13 includes the display device of example 1, wherein at least one of the sensor is to generate the detection signal or the logic circuitry is to generate the control signal in response to an enable signal from host hardware in communication with the display device. [0240] Example 14 includes the display device of example 1, wherein at least one of the sensor or the logic circuitry is to be disabled in response to a disable signal from host hardware in communication with the display device. [0241] Example 15 includes the display device of example 14, wherein after the disabling of the at least one of the sensor or the logic circuitry, the at least one of the sensor or the logic circuitry is to be enabled in response to an enable signal from the host hardware. [0242] Example 16 includes the display device of example 1, wherein at least one of the sensor or the logic circuitry is to maintain operation in response to a disconnection between the display device and host hardware. [0243] Example 17 includes a computer readable storage medium comprising instructions that, when executed, cause a display device to at least generate an alert indicative of a change in human presence proximate the display device, the alert based on an output from a sensor included in the display device, detect that the display device is in communication with host hardware, and in response to the alert and the detection, adjust a power state of the display device. [0244] Example 18 includes the computer readable storage medium of example 17, wherein the instructions, when executed, cause the display device to, in response to the human presence and the detection, adjust the power state by increasing a backlight intensity of the display device or turning on the display device. [0245] Example 19 includes the computer readable storage medium of example 17, wherein the instructions, when executed, cause the display device to, in response to human absence and the detection, adjust the power state by decreasing a backlight intensity of the display device or turning off the display device. [0246] Example 20 includes the computer readable storage medium of example 17, wherein the instructions, when executed, cause the display device to generate a connection status signal in response to detecting that the display device is in communication with the host hardware, in response to alert and the connection status signal, trigger a timer, in response to the timer satisfying a threshold without the alert indicative of a presence of a human during a time period associated with the timer, turn off the display device, and transmit a control signal to the host hardware to cause the host hardware to stop transmission of display data from the host hardware to the display device. [0247] Example 21 includes the computer readable storage medium of example 17, wherein the instructions, when executed, cause the display device to configure the sensor based on detection parameters, obtain the alert from the sensor, and transmit at least one of the detection parameters or the alert to the host hardware. [0248] Example 22 includes the computer readable storage medium of example 21, wherein the instructions, when executed, cause the display device to compare the at least one of the detection parameters or the alert to a power management profile, determine to adjust the power state of the display device based on the comparison, and invoke the display device to change the power state based on the determination. [0249] Example 23 includes the computer readable storage medium of example 17, wherein the instructions, when executed, cause the display device to cause the sensor to generate the output in response to an enable signal from the host hardware. [0250] Example 24 includes the computer readable storage medium of example 17, wherein the instructions, when executed, cause the display device to disable the sensor in response to a disable signal from the host hardware. [0251] Example 25 includes the computer readable storage medium of example 24, wherein after the disabling of the sensor, the instructions, when executed, cause the display device to enable the sensor in

response to an enable signal from the host hardware. [0252] Example 26 includes the computer readable storage medium of example 17, wherein the instructions, when executed, cause the display device to cause the sensor to maintain operation in response to a disconnection between the display device and host hardware. [0253] Example 27 includes a display device comprising first means for generating a detection signal based on whether a human is detected, and second means for generating a control signal to cause an adjustment of a power state of the display device based on the detection signal. [0254] Example 28 includes the display device of example 27, wherein the second means for generating is to generate the control signal to reduce a backlight intensity of the display device or turn off the display device. [0255] Example 29 includes the display device of example 27, wherein the second means for generating is to determine that the detection signal is indicative of a presence of the human proximate the display device, and determine that a connection status signal indicates the display device is in communication with host hardware. [0256] Example 30 includes the display device of example 27, wherein the control signal is to adjust the power state of the display device from a disabled state to an enabled state. [0257] Example 31 includes the display device of example 27, wherein the second means for generating is to determine that the detection signal is indicative of an absence of the human proximate the display device, and determine that a connection status signal indicates the display device is in communication with host hardware. [0258] Example 32 includes the display device of example 31, wherein the control signal is to adjust the power state of the display device from an enabled state to a disabled state. [0259] Example 33 includes the display device of example 27, further including means for asserting a connection status signal in response to detecting that the display device is in communication with host hardware, and means for de-asserting the connection status signal in response to not detecting that the display device is in communication with the host hardware. [0260] Example 34 includes the display device of example 27, wherein the control signal is a first control signal, and further including third means for generating a connection status signal in response to detecting that the display device is in communication with host hardware, in response to obtaining the control signal, means for triggering a timer, in response to the timer elapsing, means for turning off the display device, and means for transmitting a second control signal to the host hardware to cease transmission of display data from the host hardware to the display device. [0261] Example 35 includes the display device of example 34, wherein the means for turning is to turn off the display device in response to the timer elapsing without the detection signal changing indication of an absence of the human. [0262] Example 36 includes the display device of example 27, further including means for configuring the first means for generating based on detection parameters, and means for transmitting at least one of the detection parameters or the detection signal to host hardware in communication with the display device. [0263] Example 37 includes the display device of example 27, wherein the first means for generating is to generate the detection signal based on whether the human is detected proximate the display device. [0264] Example 38 includes the display device of example 27, wherein at least one of the first means for generating is to generate the detection signal or the second means for generating is to generate the control signal in response to an enable signal from host hardware in communication with the display device. [0265] Example 39 includes the display device of example 27, wherein at least one of the first means for generating or the second means for generating is to be disabled in response to a disable signal from host hardware in communication with the display device. [0266] Example 40 includes the display device of example 39, wherein after the disabling of the at least one of the first means for generating or the second means for generating, the at least one of the first means for generating or the second means for generating is to be enabled in response to an enable signal from the host hardware. [0267] Example 41 includes the display device of example 27, wherein at least one of the first means for generating or the second means for generating is to maintain operation in response to a disconnection between the display device and host hardware. [0268] Example 42 includes a method for human presence detection, the method comprising generating an alert

indicative of a change in human presence proximate a display device, the alert based on an output from a sensor included in the display device, detecting that the display device is in communication with host hardware, and in response to the alert and the detection, adjusting a power state of the display device. [0269] Example 43 includes the method of example 42, further including, in response to the human presence and the detection, adjusting the power state by increasing a backlight intensity of the display device or turning on the display device. [0270] Example 44 includes the method of example 42, further including, in response to human absence and the detection, adjusting the power state by decreasing a backlight intensity of the display device or turning off the display device. [0271] Example 45 includes the method of example 42, further including generating a connection status signal in response to detecting that the display device is in communication with the host hardware, in response to alert and the connection status signal, triggering a timer, in response to the timer satisfying a threshold without the alert indicative of a presence of a human during a time period associated with the timer, turning off the display device, and transmitting a control signal to the host hardware to cause the host hardware to stop transmission of display data from the host hardware to the display device. [0272] Example 46 includes the method of example 42, further including configuring the sensor based on detection parameters, obtaining the alert from the sensor, and transmitting at least one of the detection parameters or the alert to the host hardware. [0273] Example 47 includes the method of example 46, further including comparing the at least one of the detection parameters or the alert to a power management profile, determining to adjust the power state of the display device based on the comparison, and invoking the display device to change the power state based on the determination. [0274] Example 48 includes the method of example 42, further including enabling the sensor in response to an enable signal from the host hardware. [0275] Example 49 includes the method of example 42, further including disabling the sensor in response to a disable signal from the host hardware. [0276] Example 50 includes the method of example 49, further including after the disabling of the sensor, enabling the sensor in response to an enable signal from the host hardware. [0277] Example 51 includes the method of example 42, wherein the sensor is to maintain operation in response to a disconnection between the display device and host hardware. [0278] Example 52 includes the method of example 42, further including querying, by the host hardware, the display device for whether the sensor is included in the display device, and in response to the display device not including the sensor, transmitting, from the host hardware, display data to the display device. [0279] Example 53 includes the method of example 42, wherein the display device is a first display device, and further including in response to a disconnection between the first display device and the host hardware, maintain operation of the host hardware, and in response to a connection between a second display device and the host hardware, transmit display data to the second display device, the second display device not including the sensor. [0280] Although certain example systems, methods, apparatus, and articles of manufacture have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, methods, apparatus, and articles of manufacture fairly falling within the scope of the claims of this patent.

Claims

1. (canceled)
2. A display device comprising: a sensor to detect a presence of a human; at least one logic circuit to: determine a connection status indicative of whether the display device is connected to host hardware; trigger a timer when the sensor does not detect a human; after the timer elapses, cause the display device to switch from a first power state to a second power state, the second power state lower than the first power state; generate a control signal based on (a) the connection status and (b) whether the sensor detects a human; and cause transmission of the control signal to the host

hardware.

3. The display device of claim 2, wherein one or more of the at least one logic circuit is to, after the timer elapses, change the connection status to indicate the display device is not connected to the host hardware.
4. The display device of claim 2, wherein the control signal is to be provided to the host hardware via an auxiliary channel between the display device and the host hardware.
5. The display device of claim 2, wherein the second power state is an off state of the display device.
6. The display device of claim 2, wherein the first power state is associated with a first brightness and the second power state is associated with a second brightness, the second brightness different from the first brightness.
7. The display device of claim 2, wherein the control signal is to cause the host hardware to cease transmission of display data to the display device.
8. The display device of claim 2, wherein one or more of the at least one logic circuit is to cause the display device to switch from a third power state to the first power state when the sensor does not detect a human and the timer is triggered, the first power state lower than the third power state.
9. A non-transitory machine readable storage medium comprising instructions to cause at least one logic circuit to at least: detect a presence of a human based on a signal from a sensor of a display; determine whether the display is connected to host hardware; trigger a timer when the sensor does not detect a human; after elapsing of the timer, cause the display device to switch from a first power state to a second power state, the second power state lower than the first power state; generate a control signal based on (a) the determination and (b) whether the sensor detects a human; and cause transmission of the control signal to the host hardware.
10. The non-transitory machine readable storage medium of claim 9, wherein the instructions are to cause one or more of the at least one logic circuit to, after the elapsing of the timer, change the determination to indicate the display is not connected to the host hardware.
11. The non-transitory machine readable storage medium of claim 9, wherein the control signal is to be provided to the host hardware via an auxiliary channel between the display and the host hardware.
12. The non-transitory machine readable storage medium of claim 9, wherein the second power state is an off state of the display.
13. The non-transitory machine readable storage medium of claim 9, wherein the first power state is associated with a first brightness and the second power state is associated with a second brightness, the second brightness different from the first brightness.
14. The non-transitory machine readable storage medium of claim 9, wherein the control signal is to cause the host hardware to cease transmission of display data to the display.
15. The non-transitory machine readable storage medium of claim 9, wherein the instructions are to cause the one or more of the at least one logic circuit to cause the display to switch from a third power state to the first power state when the sensor does not detect a human and the timer is triggered, the first power state lower than the third power state.
16. A method comprising: detecting, with a sensor of a display device, a presence of a human using a sensor; determining a connection status indicative of whether the display device is connected to host hardware; triggering a timer when the sensor does not detect a human; after the timer elapses, switching the display device from a first power state to a second power state, the second power state lower than the first power state; generating, by executing an instruction with at least one logic circuit, a control signal based on (a) the connection status and (b) whether the sensor detects a human; and transmitting of the control signal to the host hardware.
17. The method of claim 16, including changing, after the timer elapses, the connection status to indicate the display device is not connected to the host hardware.
18. The method of claim 16, wherein the control signal is to be provided to the host hardware via

an auxiliary channel between the display device and the host hardware.

19. The method of claim 16, wherein the second power state is an off state of the display device.

20. The method of claim 16, wherein the first power state is associated with a first brightness and the second power state is associated with a second brightness, the second brightness different from the first brightness.

21. The method of claim 16, wherein the control signal is to cause the host hardware to cease transmission of display data to the display device.
